

Class 1 Homework

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

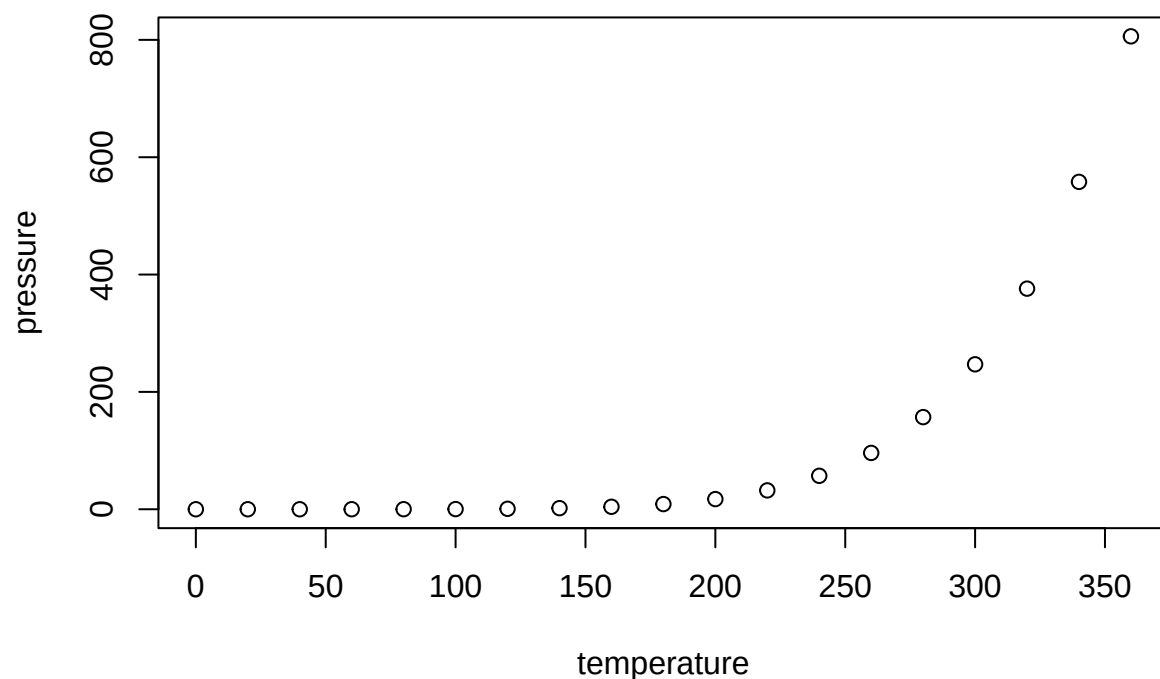
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0    Min.   :  2.00
##  1st Qu.:12.0    1st Qu.: 26.00
##  Median :15.0    Median : 36.00
##  Mean   :15.4    Mean   : 42.98
##  3rd Qu.:19.0    3rd Qu.: 56.00
##  Max.   :25.0    Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

```
knitr::opts_chunk$set(cache.extra = knitr::rand_seed)
```

```
tryCatch(require(pacman),finally=utils::install.packages(pkgs='pacman',repos='http://cran.r-project.org'))
```

```
## Loading required package: pacman
```

```
##
```

```
## The downloaded binary packages are in
```

```
## /var/folders/cp/2n7bsp3s4p3c4t2zy2vpcnn00000gn/T//Rtmp1Km0aU/downloaded_packages
```

```
require(pacman)
```

```
pacman::p_load(Hmisc,  
               checkmate,  
               corrr,  
               conflicted,  
               readxl,  
               dplyr,  
               tidyr,  
               ggplot2,  
               knitr,  
               evaluate,  
               iopsych,  
               psych,  
               quantreg,
```

```

    lavaan,
    xtable,
    reshape2,
    GPArotation,
    Amelia,
    # esquisse,
    expss,
    multilevel,
    janitor,
    mice,
    lmtest,
    naniar,
    tidylog,
    haven
  )

## Warning: package 'iopsyh' is not available for this version of R
##
## A version of this package for your version of R might be available elsewhere,
## see the ideas at
## https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#Installing-packages
##
## Warning: 'BiocManager' not available. Could not check Bioconductor.
##
## Please use `install.packages('BiocManager')` and then retry.
##
## Warning in p_install(package, character.only = TRUE, ...):
##
## Warning in library(package, lib.loc = lib.loc, character.only = TRUE,
## logical.return = TRUE, : there is no package called 'iopsyh'
##
## Warning in pacman::p_load(Hmisc, checkmate, corrr, conflicted, readxl, dplyr, : Failed to install/load
## iopsyh

```

Loading Data

Now we're going to load the data and take a look at it.

```

#' <!-- #####
#' <!-- #####
#' <!-- #####LOADING DATA#####
library(psych)
library(haven)

#' Using the preexisting data set in the `psych` package
df2 <- read_sav("SAQ.sav")

df2 <- haven::zap_labels(df2)
df2 <- haven::zap_formats(df2)
df2 <- haven::zap_widths(df2)

str(df2)

## tibble [2,571 x 31] (S3: tbl_df/tbl/data.frame)
## $ Question_01: num [1:2571] 2 1 2 3 2 2 2 3 2 ...
## ..- attr(*, "label")= chr "Statistics makes me cry"

```

```

## $ Question_02: num [1:2571] 1 1 3 1 1 1 3 2 3 4 ...
##   .- attr(*, "label")= chr "My friends will think I'm stupid for not being able to cope with SPSS"
## $ Question_03: num [1:2571] 4 4 2 1 3 3 3 3 1 4 ...
##   .- attr(*, "label")= chr "Standard deviations excite me"
## $ Question_04: num [1:2571] 2 3 2 4 2 2 2 2 4 3 ...
##   .- attr(*, "label")= chr "I dream that Pearson is attacking me with correlation coefficients"
## $ Question_05: num [1:2571] 2 2 4 3 2 4 2 2 5 2 ...
##   .- attr(*, "label")= chr "I don't understand statistics"
## $ Question_06: num [1:2571] 2 2 1 3 3 4 2 2 3 1 ...
##   .- attr(*, "label")= chr "I have little experience of computers"
## $ Question_07: num [1:2571] 3 2 2 4 3 4 2 2 5 2 ...
##   .- attr(*, "label")= chr "All computers hate me"
## $ Question_08: num [1:2571] 1 2 2 2 2 2 2 2 5 2 ...
##   .- attr(*, "label")= chr "I have never been good at mathematics"
## $ Question_09: num [1:2571] 1 5 2 2 4 4 3 4 3 3 ...
##   .- attr(*, "label")= chr "My friends are better at statistics than me"
## $ Question_10: num [1:2571] 2 2 2 4 2 3 2 2 3 2 ...
##   .- attr(*, "label")= chr "Computers are useful only for playing games"
## $ Question_11: num [1:2571] 1 2 3 2 2 2 2 2 5 2 ...
##   .- attr(*, "label")= chr "I did badly at mathematics at school"
## $ Question_12: num [1:2571] 2 3 3 2 3 4 2 3 5 3 ...
##   .- attr(*, "label")= chr "People try to tell you that SPSS makes statistics easier to understand"
## $ Question_13: num [1:2571] 2 1 2 2 3 3 2 2 5 2 ...
##   .- attr(*, "label")= chr "I worry that I will cause irreparable damage because of my incompetence"
## $ Question_14: num [1:2571] 2 3 4 3 2 3 2 2 5 1 ...
##   .- attr(*, "label")= chr "Computers have minds of their own and deliberately go wrong whenever I use them"
## $ Question_15: num [1:2571] 2 4 2 3 2 5 2 3 5 2 ...
##   .- attr(*, "label")= chr "Computers are out to get me"
## $ Question_16: num [1:2571] 3 3 3 3 2 2 2 2 5 3 ...
##   .- attr(*, "label")= chr "I weep openly at the mention of central tendency"
## $ Question_17: num [1:2571] 1 2 2 2 2 3 2 2 5 2 ...
##   .- attr(*, "label")= chr "I slip into a coma whenever I see an equation"
## $ Question_18: num [1:2571] 2 2 3 4 3 5 2 2 5 2 ...
##   .- attr(*, "label")= chr "SPSS always crashes when I try to use it"
## $ Question_19: num [1:2571] 3 3 1 2 3 1 3 4 2 3 ...
##   .- attr(*, "label")= chr "Everybody looks at me when I use SPSS"
## $ Question_20: num [1:2571] 2 4 4 4 4 5 2 3 5 3 ...
##   .- attr(*, "label")= chr "I can't sleep for thoughts of eigenvectors"
## $ Question_21: num [1:2571] 2 4 3 4 2 3 2 2 5 2 ...
##   .- attr(*, "label")= chr "I wake up under my duvet thinking that I am trapped under a normal distribution"
## $ Question_22: num [1:2571] 2 4 2 4 4 1 4 4 3 4 ...
##   .- attr(*, "label")= chr "My friends are better at SPSS than I am"
## $ Question_23: num [1:2571] 5 2 2 3 4 4 4 4 3 4 ...
##   .- attr(*, "label")= chr "If I'm good at statistics my friends will think I'm a nerd"
## $ FAC1_1      : num [1:2571] -1.1297 -0.0448 0.1562 0.7937 -0.9825 ...
##   .- attr(*, "label")= chr "A-R factor score 1 for analysis 1"
## $ FAC2_1      : num [1:2571] 0.0509 -0.4774 -0.7224 0.6118 0.6628 ...
##   .- attr(*, "label")= chr "A-R factor score 2 for analysis 1"
## $ FAC3_1      : num [1:2571] -1.586 -0.221 0.083 -0.793 -0.358 ...
##   .- attr(*, "label")= chr "A-R factor score 3 for analysis 1"
## $ FAC4_1      : num [1:2571] -0.552 0.641 -0.909 -0.318 0.548 ...
##   .- attr(*, "label")= chr "A-R factor score 4 for analysis 1"
## $ FAC1_2      : num [1:2571] -0.9452 -0.0171 0.2749 0.8158 -1.0391 ...
##   .- attr(*, "label")= chr "A-R factor score 1 for analysis 2"

```

```
## $ FAC2_2 : num [1:2571] -0.618 0.61 -0.934 -0.227 0.516 ...
## .. attr(*, "label")= chr "A-R factor score 2 for analysis 2"
## $ FAC3_2 : num [1:2571] 0.0378 -0.5085 -0.6518 0.7289 0.5483 ...
## .. attr(*, "label")= chr "A-R factor score 3 for analysis 2"
## $ FAC4_2 : num [1:2571] 1.6807 0.2401 -0.0804 0.6975 0.4316 ...
## .. attr(*, "label")= chr "A-R factor score 4 for analysis 2"
```

Glimpse the data.

```
glimpse(df2)
```

```
## Rows: 2,571
## Columns: 31
## $ Question_01 <dbl> 2, 1, 2, 3, 2, 2, 2, 2, 3, 2, 2, 2, 3, 2, 2, 3, 1, 2, 2, 2~
## $ Question_02 <dbl> 1, 1, 3, 1, 1, 1, 3, 2, 3, 4, 1, 1, 1, 2, 2, 1, 2, 2, 3, 1~
## $ Question_03 <dbl> 4, 4, 2, 1, 3, 3, 3, 3, 1, 4, 5, 3, 3, 1, 3, 2, 5, 3, 4, 1~
## $ Question_04 <dbl> 2, 3, 2, 4, 2, 2, 2, 2, 4, 3, 2, 3, 4, 2, 4, 2, 2, 3, 2, 2~
## $ Question_05 <dbl> 2, 2, 4, 3, 2, 4, 2, 2, 5, 2, 2, 4, 3, 2, 2, 2, 1, 3, 3, 3~
## $ Question_06 <dbl> 2, 2, 1, 3, 3, 4, 2, 2, 3, 1, 1, 3, 2, 2, 2, 2, 1, 4, 1, 4~
## $ Question_07 <dbl> 3, 2, 2, 4, 3, 4, 2, 2, 5, 2, 2, 3, 3, 3, 3, 2, 1, 3, 1, 4~
## $ Question_08 <dbl> 1, 2, 2, 2, 2, 2, 2, 2, 5, 2, 2, 1, 3, 2, 2, 2, 1, 2, 1, 1~
## $ Question_09 <dbl> 1, 5, 2, 2, 4, 4, 3, 4, 3, 3, 5, 3, 2, 2, 2, 2, 4, 5, 5, 5~
## $ Question_10 <dbl> 2, 2, 2, 4, 2, 3, 2, 2, 3, 2, 2, 2, 3, 3, 3, 3, 1, 2, 2, 1~
## $ Question_11 <dbl> 1, 2, 3, 2, 2, 2, 2, 2, 5, 2, 1, 2, 3, 2, 2, 2, 1, 3, 1, 2~
## $ Question_12 <dbl> 2, 3, 3, 2, 3, 4, 2, 3, 5, 3, 3, 3, 4, 4, 3, 3, 2, 3, 3, 5~
## $ Question_13 <dbl> 2, 1, 2, 2, 3, 3, 2, 2, 5, 2, 1, 2, 4, 2, 2, 2, 1, 3, 1, 2~
## $ Question_14 <dbl> 2, 3, 4, 3, 2, 3, 2, 2, 5, 1, 2, 2, 4, 4, 3, 3, 1, 3, 2, 5~
## $ Question_15 <dbl> 2, 4, 2, 3, 2, 5, 2, 3, 5, 2, 1, 3, 4, 4, 3, 2, 1, 4, 2, 5~
## $ Question_16 <dbl> 3, 3, 3, 3, 2, 2, 2, 2, 5, 3, 2, 3, 4, 4, 4, 3, 2, 3, 3, 5~
## $ Question_17 <dbl> 1, 2, 2, 2, 2, 3, 2, 2, 5, 2, 2, 2, 3, 2, 2, 2, 2, 2, 1, 2~
## $ Question_18 <dbl> 2, 2, 3, 4, 3, 5, 2, 2, 5, 2, 2, 2, 3, 4, 3, 3, 1, 2, 1, 5~
## $ Question_19 <dbl> 3, 3, 1, 2, 3, 1, 3, 4, 2, 3, 5, 3, 2, 1, 3, 2, 4, 2, 4, 1~
## $ Question_20 <dbl> 2, 4, 4, 4, 4, 5, 2, 3, 5, 3, 3, 4, 4, 5, 4, 3, 2, 3, 2, 5~
## $ Question_21 <dbl> 2, 4, 3, 4, 2, 3, 2, 2, 5, 2, 2, 3, 4, 5, 4, 2, 1, 3, 2, 5~
## $ Question_22 <dbl> 2, 4, 2, 4, 4, 1, 4, 4, 3, 4, 5, 4, 3, 3, 4, 3, 4, 3, 4, 5~
## $ Question_23 <dbl> 5, 2, 2, 3, 4, 4, 4, 4, 3, 4, 5, 4, 4, 1, 4, 4, 4, 4, 4, 5~
## $ FAC1_1 <dbl> -1.12973669, -0.04484331, 0.15619586, 0.79369971, -0.98250~
## $ FAC2_1 <dbl> 0.05090334, -0.47738650, -0.72239801, 0.61177764, 0.662841~
## $ FAC3_1 <dbl> -1.58646169, -0.22125910, 0.08298662, -0.79340882, -0.3581~
## $ FAC4_1 <dbl> -0.55242061, 0.64055340, -0.90900640, -0.31779452, 0.54788~
## $ FAC1_2 <dbl> -0.94520232, -0.01713883, 0.27492721, 0.81581663, -1.03907~
## $ FAC2_2 <dbl> -0.61761674, 0.61044497, -0.93408252, -0.22690610, 0.51633~
## $ FAC3_2 <dbl> 0.03775347, -0.50851122, -0.65182212, 0.72885731, 0.548270~
## $ FAC4_2 <dbl> 1.680667173, 0.240109522, -0.080425498, 0.697456742, 0.431~
```

Get column names.

```
colnames(df2)
```

```
## [1] "Question_01" "Question_02" "Question_03" "Question_04" "Question_05"
## [6] "Question_06" "Question_07" "Question_08" "Question_09" "Question_10"
## [11] "Question_11" "Question_12" "Question_13" "Question_14" "Question_15"
## [16] "Question_16" "Question_17" "Question_18" "Question_19" "Question_20"
## [21] "Question_21" "Question_22" "Question_23" "FAC1_1" "FAC2_1"
## [26] "FAC3_1" "FAC4_1" "FAC1_2" "FAC2_2" "FAC3_2"
## [31] "FAC4_2"
```

Get column names wrapped in the combine function, c().

```
dput(colnames(df2))
```

```
## c("Question_01", "Question_02", "Question_03", "Question_04",  
## "Question_05", "Question_06", "Question_07", "Question_08", "Question_09",  
## "Question_10", "Question_11", "Question_12", "Question_13", "Question_14",  
## "Question_15", "Question_16", "Question_17", "Question_18", "Question_19",  
## "Question_20", "Question_21", "Question_22", "Question_23", "FAC1_1",  
## "FAC2_1", "FAC3_1", "FAC4_1", "FAC1_2", "FAC2_2", "FAC3_2", "FAC4_2"  
## )
```

Exploratory df Analysis (EDA)

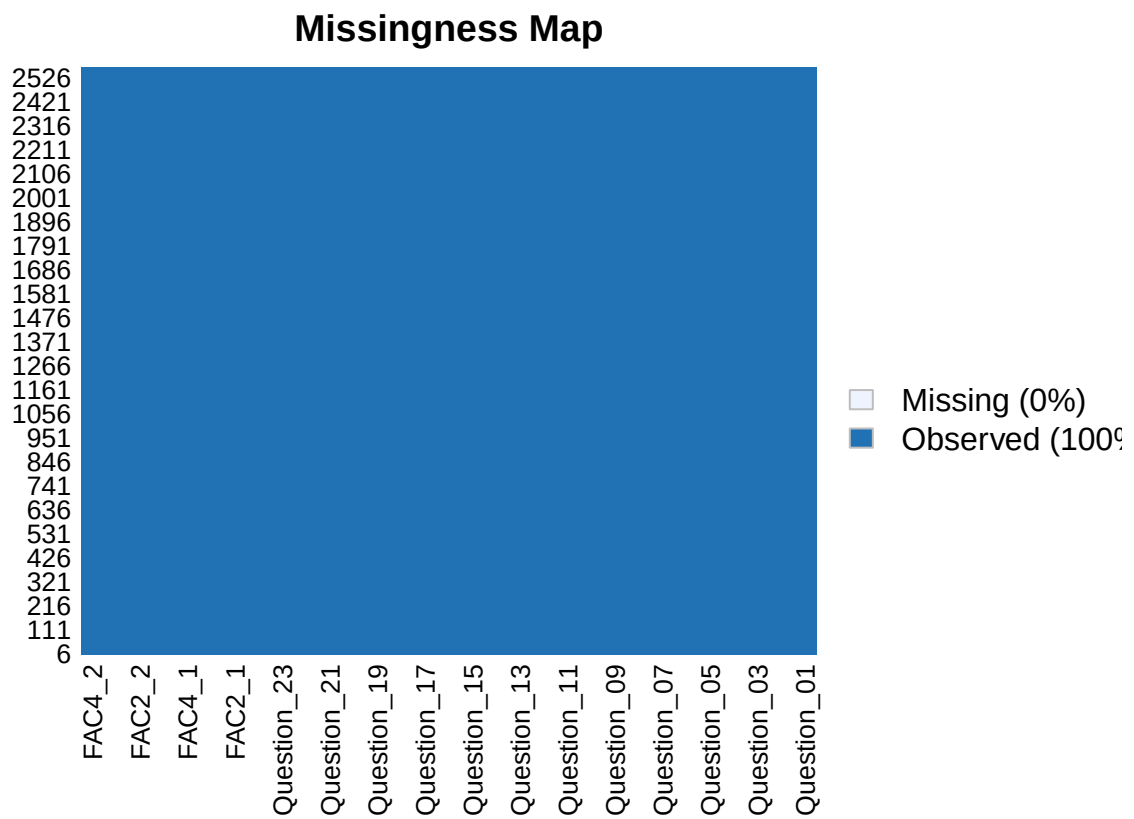
Missing Data

Now we're going to look for missing data using the *Amelia* package.

```
## Looking for missing data  
library(Amelia)
```

```
missmap(df2)
```

```
## Warning: Unknown or uninitialised column: `arguments`.  
## Unknown or uninitialised column: `arguments`.  
## Warning: Unknown or uninitialised column: `imputations`.
```



There is no missing data so there is no need to remove any of it.

Checking for Outliers

```
df_23 <- df2[,1:23]      #first I will remove the factor scores to get just the items

# Set the random seed for reproducibility
set.seed(2026)

# Compute the chi-square cutoff for identifying multivariate outliers.
# p = .999 means we're using the 99.9th percentile as our threshold (very conservative).
# df = number of variables in the dataset, because Mahalanobis distance follows a
# chi-square distribution with degrees of freedom equal to the number of variables.
cutoff <- qchisq(
  p = .999,
  df = ncol(df_23)
)

# Calculate the Mahalanobis distance for each observation in the dataset.
# This measures how many multivariate standard deviations each case is from
# the centroid (multivariate mean), accounting for correlations among variables.
# - colMeans(df2): the multivariate mean (centroid) used as the reference point
# - cov(df2): the covariance matrix, which accounts for variable scale and correlation
mahal = mahalanobis(df_23,
                    colMeans(df_23),
                    cov(df_23))

# Print the cutoff value so you can inspect the threshold being applied
cutoff ##cutoff score

## [1] 49.72823

# Print the degrees of freedom (number of variables) used in the chi-square distribution
ncol(df_23) ##df

## [1] 23

# Evaluate which cases fall BELOW the cutoff (i.e., are NOT outliers).
# TRUE = within acceptable distance from centroid (keep)
# FALSE = exceeds cutoff, flagged as a multivariate outlier (review/remove)
summary(mahal < cutoff)

##      Mode   FALSE    TRUE
## logical      97    2474
```

In this case FALSE is bad.

Let's take a look at those labeled outliers.

```
df_mahal <- df_23 %>%
  bind_cols(mahal) %>%
  rename(mahal = `...24`) # renaming the new column "mahal"
```

```
## New names:
## rename: renamed one variable (mahal)
## * `` -> `...24`
```

Take a look at the 97 flagged as "FALSE".

```
mahal_out <- df_mahal %>%
  filter(mahal > cutoff) %>%
  arrange(desc(mahal)) # sort mahal values from most to least
```

```
## filter: removed 2,474 rows (96%), 97 rows remaining
```

After reviewing the outlier data, I've decided to not remove them as I could not definitively determine if they were invalid cases.

Now I will reverse score items that need to be reverse scored.

```
df2 <- df2 %>%      # I am reverse scoring these items as they are asked differently from the others
  mutate(Question_02 = 6-Question_02,
         Question_03 = 6-Question_03,
         Question_19 = 6-Question_19)
```

```
## mutate: changed 2,365 values (92%) of 'Question_02' (0 new NAs)
```

```
##          changed 1,693 values (66%) of 'Question_03' (0 new NAs)
```

```
##          changed 2,015 values (78%) of 'Question_19' (0 new NAs)
```

I will now take a look at the Additivity.

```
#additivity
correl = cor(df2, use = "pairwise.complete.obs") #I am using df2 again as I chose not to remove any of
symnum(correl)
```

```
##          Q_01 Q_02 Q_03 Q_04 Q_05 Q_06 Q_07 Q_08 Q_09 Q_10 Q_11 Q_12 Q_13
## Question_01 1
## Question_02      1
## Question_03 .      .      1
## Question_04 .      .      .      1
## Question_05 .      .      .      .      1
## Question_06      .      .      .      .      .      1
## Question_07 .      .      .      .      .      .      1
## Question_08 .      .      .      .      .      .      .      1
## Question_09      .      .      .      .      .      .      .      .      1
## Question_10      .      .      .      .      .      .      .      .      .      1
## Question_11 .      .      .      .      .      .      .      .      .      .      1
## Question_12 .      .      .      .      .      .      .      .      .      .      .      1
## Question_13 .      .      .      .      .      .      .      .      .      .      .      .      1
## Question_14 .      .      .      .      .      .      .      .      .      .      .      .      .
## Question_15 .      .      .      .      .      .      .      .      .      .      .      .      .
## Question_16 .      .      .      .      .      .      .      .      .      .      .      .      .
## Question_17 .      .      .      .      .      .      .      .      .      .      .      .      .
## Question_18 .      .      .      .      .      .      .      .      .      .      .      .      .
## Question_19      .
## Question_20      .
## Question_21 .      .      .      .      .      .      .      .      .      .      .      .      .
## Question_22
## Question_23
## FAC1_1      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC2_1      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC3_1      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC4_1      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC1_2      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC2_2      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC3_2      .      .      .      .      .      .      .      .      .      .      .      .      .
## FAC4_2      .      .      .      .      .      .      .      .      .      .      .      .      .
##          Q_14 Q_15 Q_16 Q_17 Q_18 Q_19 Q_20 Q_21 Q_22 Q_23 FAC1_1 FAC2_1
## Question_01
```



```

## Question_02
## Question_03
## Question_04
## Question_05
## Question_06
## Question_07
## Question_08
## Question_09
## Question_10
## Question_11
## Question_12
## Question_13
## Question_14 1
## Question_15 . 1
## Question_16 . . 1
## Question_17 . . . 1
## Question_18 . . . . 1
## Question_19 . . . . . 1
## Question_20 . . . . . . 1
## Question_21 . . . . . . . 1
## Question_22 . . . . . . . . 1
## Question_23 . . . . . . . . . 1
## FAC1_1 . . . . . . . . . 1
## FAC2_1 . . . . . . . . . . 1
## FAC3_1 . . . . . . . . . . .
## FAC4_1 . . . . . . . . . . .
## FAC1_2 . . . . . . . . . . B
## FAC2_2 . . . . . . . . . . .
## FAC3_2 . . . . . . . . . . . B
## FAC4_2 . . . . . . . . . . .
## FAC3_1 FAC4_1 FAC1_2 FAC2_2 FAC3_2 FAC4_2
## Question_01
## Question_02
## Question_03
## Question_04
## Question_05
## Question_06
## Question_07
## Question_08
## Question_09
## Question_10
## Question_11
## Question_12
## Question_13
## Question_14
## Question_15
## Question_16
## Question_17
## Question_18
## Question_19
## Question_20
## Question_21
## Question_22
## Question_23

```

```

## FAC1_1
## FAC2_1
## FAC3_1      1
## FAC4_1      1
## FAC1_2      1
## FAC2_2      B      1
## FAC3_2      1
## FAC4_2      B      1
## attr("legend")
## [1] 0 ' ' 0.3 '.' 0.6 ',' 0.8 '+' 0.9 '*' 0.95 'B' 1

```

```
correl
```

```

##      Question_01 Question_02 Question_03 Question_04 Question_05
## Question_01 1.000000000 0.0987240320 0.3366489 0.43586018 0.402439917
## Question_02 0.098724032 1.000000000 0.3183902 0.11185965 0.119346578
## Question_03 0.336648879 0.3183901974 1.0000000 0.38046016 0.310308789
## Question_04 0.435860179 0.1118596539 0.3804602 1.00000000 0.400672247
## Question_05 0.402439917 0.1193465777 0.3103088 0.40067225 1.000000000
## Question_06 0.216733985 0.0742096790 0.2267405 0.27820154 0.257460144
## Question_07 0.305365139 0.1591744831 0.3819533 0.40861502 0.339391789
## Question_08 0.330737608 0.0496225705 0.2586342 0.34942939 0.268626968
## Question_09 -0.092339458 -0.3146405397 -0.2998036 -0.12454637 -0.095701508
## Question_10 0.213681706 0.0840031575 0.1933887 0.21581010 0.258209250
## Question_11 0.356786290 0.1438298375 0.3506397 0.36865655 0.297828819
## Question_12 0.345381133 0.1948694597 0.4099513 0.44164706 0.346743248
## Question_13 0.354646283 0.1427402629 0.3179193 0.34429168 0.301821588
## Question_14 0.337879655 0.1646999114 0.3707551 0.35080964 0.315338103
## Question_15 0.245752635 0.1649958134 0.3123968 0.33423089 0.261371904
## Question_16 0.498618057 0.1675522850 0.4186478 0.41586725 0.394917953
## Question_17 0.370550512 0.0869952679 0.3273715 0.38273945 0.310417218
## Question_18 0.347118037 0.1638941483 0.3752329 0.38200149 0.322091477
## Question_19 0.189011027 0.2032974792 0.3415737 0.18597751 0.165322098
## Question_20 0.213897945 0.2015943686 0.3248338 0.24291796 0.199669454
## Question_21 0.329153138 0.2046172986 0.4171878 0.41029317 0.334614937
## Question_22 -0.104408664 -0.2308748720 -0.2036569 -0.09838349 -0.132535934
## Question_23 -0.004480593 -0.0996782848 -0.1502065 -0.03381815 -0.041656841
## FAC1_1      0.585275638 0.2082643550 0.5544686 0.60494842 0.499328749
## FAC2_1      0.194674358 -0.0004466139 0.1462249 0.26913534 0.261135932
## FAC3_1      0.261508157 -0.0146205837 0.1500326 0.23831967 0.171209620
## FAC4_1      0.045876437 -0.5938957475 -0.4705470 0.02136682 -0.006694868
## FAC1_2      0.533252629 0.2443034561 0.5497381 0.54896916 0.453082270
## FAC2_2      0.094670568 -0.5775492093 -0.4240492 0.07566735 0.040408788
## FAC3_2      0.234926003 0.0547983597 0.2189467 0.31301441 0.299482542
## FAC4_2      -0.320040204 -0.0070734190 -0.2068187 -0.30096266 -0.224215351
##      Question_06 Question_07 Question_08 Question_09 Question_10
## Question_01 0.21673399 0.30536514 0.330737608 -0.09233946 0.21368171
## Question_02 0.07420968 0.15917448 0.049622570 -0.31464054 0.08400316
## Question_03 0.22674048 0.38195325 0.258634209 -0.29980362 0.19338871
## Question_04 0.27820154 0.40861502 0.349429391 -0.12454637 0.21581010
## Question_05 0.25746014 0.33939179 0.268626968 -0.09570151 0.25820925
## Question_06 1.00000000 0.51358048 0.222831755 -0.11264384 0.32223023
## Question_07 0.51358048 1.00000000 0.297496961 -0.12829828 0.28372299
## Question_08 0.22283175 0.29749696 1.000000000 0.01573316 0.15860850
## Question_09 -0.11264384 -0.12829828 0.015733161 1.00000000 -0.13418658

```

## Question_10	0.32223023	0.28372299	0.158608505	-0.13418658	1.00000000
## Question_11	0.32807072	0.34474770	0.629297683	-0.11552479	0.27143657
## Question_12	0.31250937	0.42298591	0.251985818	-0.16739436	0.24582591
## Question_13	0.46640487	0.44211926	0.314247164	-0.16743882	0.30196707
## Question_14	0.40224407	0.44070276	0.280589581	-0.12150197	0.25468730
## Question_15	0.35989309	0.39136675	0.299686000	-0.18657099	0.29523438
## Question_16	0.24433888	0.38854534	0.321494200	-0.18886556	0.29058576
## Question_17	0.28226121	0.39074283	0.590140219	-0.03681556	0.21832214
## Question_18	0.51332164	0.50086685	0.279744327	-0.14957782	0.29250304
## Question_19	0.16675017	0.26912031	0.159476710	-0.24931170	0.12723487
## Question_20	0.10092489	0.22095420	0.175150891	-0.15864747	0.08406520
## Question_21	0.27233273	0.48300388	0.295717563	-0.13594310	0.19313633
## Question_22	-0.16513541	-0.16820488	-0.079172646	0.25684622	-0.13090831
## Question_23	-0.06868743	-0.07029016	-0.050238392	0.17077441	-0.06191796
## FAC1_1	-0.03916833	0.37037995	0.197432927	-0.10619046	0.11656564
## FAC2_1	0.88034009	0.62056699	0.126100488	-0.05388733	0.42375029
## FAC3_1	0.11392852	0.13977738	0.866859031	0.06954650	0.12953549
## FAC4_1	-0.10144449	-0.11704434	0.008249365	0.71854665	-0.13667773
## FAC1_2	-0.12963431	0.29973915	0.104256238	-0.15068020	0.07028087
## FAC2_2	-0.05700984	-0.05886644	0.024811190	0.70550563	-0.10610220
## FAC3_2	0.87270458	0.65237653	0.111151215	-0.10883856	0.43506099
## FAC4_2	-0.13466650	-0.19135786	-0.884620921	-0.05511548	-0.15210839
##	Question_11	Question_12	Question_13	Question_14	Question_15
## Question_01	0.35678629	0.34538113	0.35464628	0.33787966	0.24575263
## Question_02	0.14382984	0.19486946	0.14274026	0.16469991	0.16499581
## Question_03	0.35063969	0.40995127	0.31791928	0.37075510	0.31239678
## Question_04	0.36865655	0.44164706	0.34429168	0.35080964	0.33423089
## Question_05	0.29782882	0.34674325	0.30182159	0.31533810	0.26137190
## Question_06	0.32807072	0.31250937	0.46640487	0.40224407	0.35989309
## Question_07	0.34474770	0.42298591	0.44211926	0.44070276	0.39136675
## Question_08	0.62929768	0.25198582	0.31424716	0.28058958	0.29968600
## Question_09	-0.11552479	-0.16739436	-0.16743882	-0.12150197	-0.18657099
## Question_10	0.27143657	0.24582591	0.30196707	0.25468730	0.29523438
## Question_11	1.00000000	0.33529466	0.42316548	0.32532025	0.36482687
## Question_12	0.33529466	1.00000000	0.48871303	0.43270398	0.33179910
## Question_13	0.42316548	0.48871303	1.00000000	0.44978632	0.34219704
## Question_14	0.32532025	0.43270398	0.44978632	1.00000000	0.38011484
## Question_15	0.36482687	0.33179910	0.34219704	0.38011484	1.00000000
## Question_16	0.36907763	0.40805908	0.35837775	0.41841820	0.45427861
## Question_17	0.58683495	0.33269383	0.40837657	0.35374183	0.37310235
## Question_18	0.37341373	0.49296482	0.53293713	0.49830615	0.34287045
## Question_19	0.19965203	0.26665953	0.22697105	0.25405813	0.20980230
## Question_20	0.25533736	0.29802585	0.20396327	0.22592173	0.20625622
## Question_21	0.34643407	0.44063832	0.37443078	0.39938896	0.29971557
## Question_22	-0.16198921	-0.16728557	-0.19535632	-0.16983754	-0.16790617
## Question_23	-0.08637256	-0.04642506	-0.05298304	-0.04847418	-0.06200665
## FAC1_1	0.19576679	0.57234829	0.26971933	0.41216495	0.26724603
## FAC2_1	0.26069122	0.40808347	0.62526378	0.52535002	0.39909017
## FAC3_1	0.78119947	0.07646892	0.22842461	0.12548674	0.26176914
## FAC4_1	-0.20565842	-0.13783918	-0.14549847	-0.11549367	-0.22821757
## FAC1_2	0.11033806	0.52748112	0.19350326	0.35166206	0.21554009
## FAC2_2	-0.18085022	-0.07704673	-0.09413291	-0.05953819	-0.18896373
## FAC3_2	0.26062107	0.46348545	0.64622389	0.56207885	0.42506897
## FAC4_2	-0.80377105	-0.14131710	-0.27044608	-0.17836667	-0.29740762

##	Question_16	Question_17	Question_18	Question_19	Question_20
## Question_01	0.49861806	0.370550512	0.34711804	0.18901103	0.21389794
## Question_02	0.16755228	0.086995268	0.16389415	0.20329748	0.20159437
## Question_03	0.41864780	0.327371453	0.37523290	0.34157370	0.32483385
## Question_04	0.41586725	0.382739448	0.38200149	0.18597751	0.24291796
## Question_05	0.39491795	0.310417218	0.32209148	0.16532210	0.19966945
## Question_06	0.24433888	0.282261208	0.51332164	0.16675017	0.10092489
## Question_07	0.38854534	0.390742834	0.50086685	0.26912031	0.22095420
## Question_08	0.32149420	0.590140219	0.27974433	0.15947671	0.17515089
## Question_09	-0.18886556	-0.036815560	-0.14957782	-0.24931170	-0.15864747
## Question_10	0.29058576	0.218322139	0.29250304	0.12723487	0.08406520
## Question_11	0.36907763	0.586834954	0.37341373	0.19965203	0.25533736
## Question_12	0.40805908	0.332693830	0.49296482	0.26665953	0.29802585
## Question_13	0.35837775	0.408376571	0.53293713	0.22697105	0.20396327
## Question_14	0.41841820	0.353741835	0.49830615	0.25405813	0.22592173
## Question_15	0.45427861	0.373102353	0.34287045	0.20980230	0.20625622
## Question_16	1.00000000	0.409763092	0.42197911	0.26704702	0.26514025
## Question_17	0.40976309	1.000000000	0.37560681	0.16288096	0.20523013
## Question_18	0.42197911	0.375606811	1.00000000	0.25663183	0.23518040
## Question_19	0.26704702	0.162880961	0.25663183	1.00000000	0.24859386
## Question_20	0.26514025	0.205230133	0.23518040	0.24859386	1.00000000
## Question_21	0.42054273	0.363491468	0.43010427	0.27489793	0.46770448
## Question_22	-0.15579385	-0.126290659	-0.15982631	-0.23392259	-0.09970186
## Question_23	-0.08152195	-0.091672426	-0.08041698	-0.12243440	-0.03466529
## FAC1_1	0.61963489	0.291882810	0.36957511	0.29314636	0.54188320
## FAC2_1	0.25127319	0.264479231	0.68199788	0.12961315	-0.01054452
## FAC3_1	0.22850256	0.713847057	0.11160277	0.05257126	0.07490088
## FAC4_1	-0.14426472	-0.041159004	-0.11492174	-0.46008885	-0.22604794
## FAC1_2	0.57644941	0.200858415	0.29528055	0.30103661	0.54432515
## FAC2_2	-0.08926838	-0.009986535	-0.05340036	-0.43180430	-0.18941026
## FAC3_2	0.30689314	0.265863612	0.71422645	0.18133135	0.05020130
## FAC4_2	-0.29268613	-0.745177313	-0.16494676	-0.08497525	-0.12557915
##	Question_21	Question_22	Question_23	FAC1_1	FAC2_1
## Question_01	0.32915314	-0.10440866	-0.004480593	5.852756e-01	1.946744e-01
## Question_02	0.20461730	-0.23087487	-0.099678285	2.082644e-01	-4.466139e-04
## Question_03	0.41718781	-0.20365686	-0.150206522	5.544686e-01	1.462249e-01
## Question_04	0.41029317	-0.09838349	-0.033818152	6.049484e-01	2.691353e-01
## Question_05	0.33461494	-0.13253593	-0.041656841	4.993287e-01	2.611359e-01
## Question_06	0.27233273	-0.16513541	-0.068687430	-3.916833e-02	8.803401e-01
## Question_07	0.48300388	-0.16820488	-0.070290157	3.703800e-01	6.205670e-01
## Question_08	0.29571756	-0.07917265	-0.050238392	1.974329e-01	1.261005e-01
## Question_09	-0.13594310	0.25684622	0.170774410	-1.061905e-01	-5.388733e-02
## Question_10	0.19313633	-0.13090831	-0.061917956	1.165656e-01	4.237503e-01
## Question_11	0.34643407	-0.16198921	-0.086372565	1.957668e-01	2.606912e-01
## Question_12	0.44063832	-0.16728557	-0.046425059	5.723483e-01	4.080835e-01
## Question_13	0.37443078	-0.19535632	-0.052983042	2.697193e-01	6.252638e-01
## Question_14	0.39938896	-0.16983754	-0.048474181	4.121649e-01	5.253500e-01
## Question_15	0.29971557	-0.16790617	-0.062006650	2.672460e-01	3.990902e-01
## Question_16	0.42054273	-0.15579385	-0.081521950	6.196349e-01	2.512732e-01
## Question_17	0.36349147	-0.12629066	-0.091672426	2.918828e-01	2.644792e-01
## Question_18	0.43010427	-0.15982631	-0.080416984	3.695751e-01	6.819979e-01
## Question_19	0.27489793	-0.23392259	-0.122434401	2.931464e-01	1.296131e-01
## Question_20	0.46770448	-0.09970186	-0.034665293	5.418832e-01	-1.054452e-02
## Question_21	1.00000000	-0.12902148	-0.067664367	6.754673e-01	2.529319e-01

```

## Question_22 -0.12902148 1.00000000 0.230369402 2.081220e-02 -1.615108e-01
## Question_23 -0.06766437 0.23036940 1.000000000 7.733359e-02 -3.577538e-02
## FAC1_1 0.67546734 0.02081220 0.077333590 1.000000e+00 -3.520153e-14
## FAC2_1 0.25293192 -0.16151078 -0.035775381 -3.520153e-14 1.000000e+00
## FAC3_1 0.12783163 -0.07870692 -0.089225262 -1.557019e-14 7.907981e-15
## FAC4_1 -0.12912358 0.59463792 0.431830032 8.301929e-15 -2.577780e-14
## FAC1_2 0.63963022 0.00662304 0.061185915 9.890735e-01 -9.876649e-02
## FAC2_2 -0.06990866 0.58548854 0.433892995 6.765600e-02 5.357938e-02
## FAC3_2 0.31646792 -0.19119437 -0.050974508 9.208441e-02 9.932759e-01
## FAC4_2 -0.19768290 0.08313806 0.084222553 -9.314942e-02 -2.788123e-02
## FAC3_1 FAC4_1 FAC1_2 FAC2_2
## Question_01 2.615082e-01 4.587644e-02 5.332526e-01 9.467057e-02
## Question_02 -1.462058e-02 -5.938957e-01 2.443035e-01 -5.775492e-01
## Question_03 1.500326e-01 -4.705470e-01 5.497381e-01 -4.240492e-01
## Question_04 2.383197e-01 2.136682e-02 5.489692e-01 7.566735e-02
## Question_05 1.712096e-01 -6.694868e-03 4.530823e-01 4.040879e-02
## Question_06 1.139285e-01 -1.014445e-01 -1.296343e-01 -5.700984e-02
## Question_07 1.397774e-01 -1.170443e-01 2.997391e-01 -5.886644e-02
## Question_08 8.668590e-01 8.249365e-03 1.042562e-01 2.481119e-02
## Question_09 6.954650e-02 7.185467e-01 -1.506802e-01 7.055056e-01
## Question_10 1.295355e-01 -1.366777e-01 7.028087e-02 -1.061022e-01
## Question_11 7.811995e-01 -2.056584e-01 1.103381e-01 -1.808502e-01
## Question_12 7.646892e-02 -1.378392e-01 5.274811e-01 -7.704673e-02
## Question_13 2.284246e-01 -1.454985e-01 1.935033e-01 -9.413291e-02
## Question_14 1.254867e-01 -1.154937e-01 3.516621e-01 -5.953819e-02
## Question_15 2.617691e-01 -2.282176e-01 2.155401e-01 -1.889637e-01
## Question_16 2.285026e-01 -1.442647e-01 5.764494e-01 -8.926838e-02
## Question_17 7.138471e-01 -4.115900e-02 2.008584e-01 -9.986535e-03
## Question_18 1.116028e-01 -1.149217e-01 2.952805e-01 -5.340036e-02
## Question_19 5.257126e-02 -4.600889e-01 3.010366e-01 -4.318043e-01
## Question_20 7.490088e-02 -2.260479e-01 5.443252e-01 -1.894103e-01
## Question_21 1.278316e-01 -1.291236e-01 6.396302e-01 -6.990866e-02
## Question_22 -7.870692e-02 5.946379e-01 6.623040e-03 5.854885e-01
## Question_23 -8.922526e-02 4.318300e-01 6.118592e-02 4.338930e-01
## FAC1_1 -1.557019e-14 8.301929e-15 9.890735e-01 6.765600e-02
## FAC2_1 7.907981e-15 -2.577780e-14 -9.876649e-02 5.357938e-02
## FAC3_1 1.000000e+00 5.638888e-15 -9.003958e-02 -4.062064e-03
## FAC4_1 5.638888e-15 1.000000e+00 -6.222332e-02 9.962607e-01
## FAC1_2 -9.003958e-02 -6.222332e-02 1.000000e+00 1.374857e-15
## FAC2_2 -4.062064e-03 9.962607e-01 1.374857e-15 1.000000e+00
## FAC3_2 -3.667074e-02 -5.982182e-02 -1.078838e-14 2.621292e-16
## FAC4_2 -9.952546e-01 3.767275e-03 -2.564802e-16 5.186157e-15
## FAC3_2 FAC4_2
## Question_01 2.349260e-01 -3.200402e-01
## Question_02 5.479836e-02 -7.073419e-03
## Question_03 2.189467e-01 -2.068187e-01
## Question_04 3.130144e-01 -3.009627e-01
## Question_05 2.994825e-01 -2.242154e-01
## Question_06 8.727046e-01 -1.346665e-01
## Question_07 6.523765e-01 -1.913579e-01
## Question_08 1.111512e-01 -8.846209e-01
## Question_09 -1.088386e-01 -5.511548e-02
## Question_10 4.350610e-01 -1.521084e-01
## Question_11 2.606211e-01 -8.037711e-01

```

```
## Question_12 4.634855e-01 -1.413171e-01
## Question_13 6.462239e-01 -2.704461e-01
## Question_14 5.620788e-01 -1.783667e-01
## Question_15 4.250690e-01 -2.974076e-01
## Question_16 3.068931e-01 -2.926861e-01
## Question_17 2.658636e-01 -7.451773e-01
## Question_18 7.142264e-01 -1.649468e-01
## Question_19 1.813314e-01 -8.497525e-02
## Question_20 5.020130e-02 -1.255791e-01
## Question_21 3.164679e-01 -1.976829e-01
## Question_22 -1.911944e-01 8.313806e-02
## Question_23 -5.097451e-02 8.422255e-02
## FAC1_1 9.208441e-02 -9.314942e-02
## FAC2_1 9.932759e-01 -2.788123e-02
## FAC3_1 -3.667074e-02 -9.952546e-01
## FAC4_1 -5.982182e-02 3.767275e-03
## FAC1_2 -1.078838e-14 -2.564802e-16
## FAC2_2 2.621292e-16 5.186157e-15
## FAC3_2 1.000000e+00 5.437749e-15
## FAC4_2 5.437749e-15 1.000000e+00
```

There are no 1s that are off of the diagonal.

```
## correlation adequacy Bartlett's test
```

```
# Run Bartlett's Test of Sphericity on the correlation matrix.
# This tests whether the correlation matrix is significantly different from
# an identity matrix (i.e., whether variables share enough correlation to
# justify factor analysis or PCA).
#
# H0: The correlation matrix is an identity matrix (all off-diagonal r = 0 -
# variables are uncorrelated and share no common variance)
# H1: The correlation matrix is NOT an identity matrix (meaningful correlations
# exist among variables, making factor analysis appropriate)
#
# correl: a pre-computed correlation matrix of your items/variables
# n = nrow(df2): the sample size from your cleaned dataset (outliers removed),
# which is required because Bartlett's test needs N to compute
# the chi-square statistic from a correlation matrix input
coratest.bartlett(correl, n = nrow(df2))
```

```
## Warning in log(detR): NaNs produced
```

```
## $chisq
## [1] NaN
##
## $p.value
## [1] NaN
##
## $df
## [1] 465
```

This shows that we have large enough correlations for an EFA.

```
## sampling adequacy KMO test
```

```
# Run the Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy.
```

```

# KMO assesses whether the pattern of correlations among variables is compact
# enough (i.e., has enough shared variance) to yield distinct, reliable factors.
# It does this by comparing observed correlations to partial correlations -
# if partial correlations are small relative to observed correlations, variables
# likely share a common factor.
#
# cor[1:23]: subsets the correlation matrix to the first 25 variables/items.
#           Explicitly indexing avoids accidentally including non-item columns
#           (e.g., ID variables, composites, or demographic fields) that may
#           have been appended to the matrix.
#
# Output includes:
# - Overall MSA (the single headline KMO value for the full set of variables)
# - Individual MSA values for each variable, which can flag items that are
#   weakly connected to the common factor structure and may warrant removal
KMO(correl[,1:23])

```

```

## Kaiser-Meyer-Olkin factor adequacy
## Call: KMO(r = correl[, 1:23])
## Overall MSA = 0.95
## MSA for each item =
## Question_01 Question_02 Question_03 Question_04 Question_05 Question_06
##          0.94          0.90          0.97          0.97          0.98          0.87
## Question_07 Question_08 Question_09 Question_10 Question_11 Question_12
##          0.96          0.88          0.92          0.96          0.92          0.97
## Question_13 Question_14 Question_15 Question_16 Question_17 Question_18
##          0.96          0.98          0.96          0.95          0.95          0.96
## Question_19 Question_20 Question_21 Question_22 Question_23
##          0.94          0.93          0.94          0.95          0.98

```

The mean sampling adequacy is .96, which is a marvelous score and this KMO confirms they're structured well enough to factor. All of the individual values are great as well so we have no questions that we should drop.

Exploratory Factor Analysis (EFA)

I will be splitting my data 50/50 as that is what my professor did and this is my first time doing it so I will trust and follow his lead. It also helps with cross-validation.

```
#####EXPLORATORY FACTOR ANALYSIS#####
```

```
set.seed(2026) #I am making sure to set my seed within my code chunk to ensure I get the same results
```

```
df2 <- df2 %>%
  mutate(id = row_number()) #I am creating an ID variable within the data set
```

```
## mutate: new variable 'id' (integer) with 2,571 unique values and 0% NA
```

```
df2 <- df2 %>%
  dplyr::select(id, everything()) #I am moving the ID variable to the front of the data set
```

```
colnames(df2) #I am checking the columns to ensure that the ID variable is at the front of the
```

```
## [1] "id" "Question_01" "Question_02" "Question_03" "Question_04"
## [6] "Question_05" "Question_06" "Question_07" "Question_08" "Question_09"
```

```
## [11] "Question_10" "Question_11" "Question_12" "Question_13" "Question_14"
## [16] "Question_15" "Question_16" "Question_17" "Question_18" "Question_19"
## [21] "Question_20" "Question_21" "Question_22" "Question_23" "FAC1_1"
## [26] "FAC2_1"      "FAC3_1"      "FAC4_1"      "FAC1_2"      "FAC2_2"
## [31] "FAC3_2"      "FAC4_2"

set.seed(2026) #I am making sure to set my seed within my code chunk to ensure I get the same results

training <- sample(df2$id, length(df2$id)*0.5) #This is splitting the data 50/50 into training and

df2_training <- subset(df2, id %in% training) #I am creating the training set to run an EFA.
df2_test <- subset(df2, !(id %in% training)) #I am creating the training set to run a CFA.
```

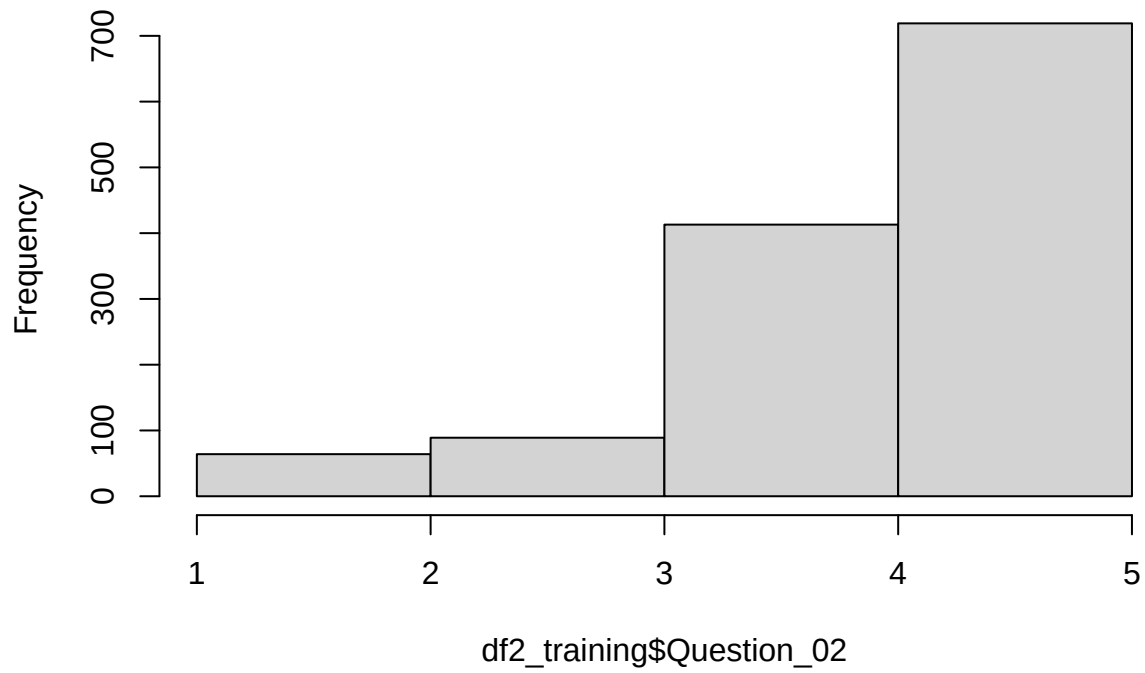
Histograms

```
#I will now take a look at the data using histograms
hist(df2_training$Question_01, breaks = 5)
```



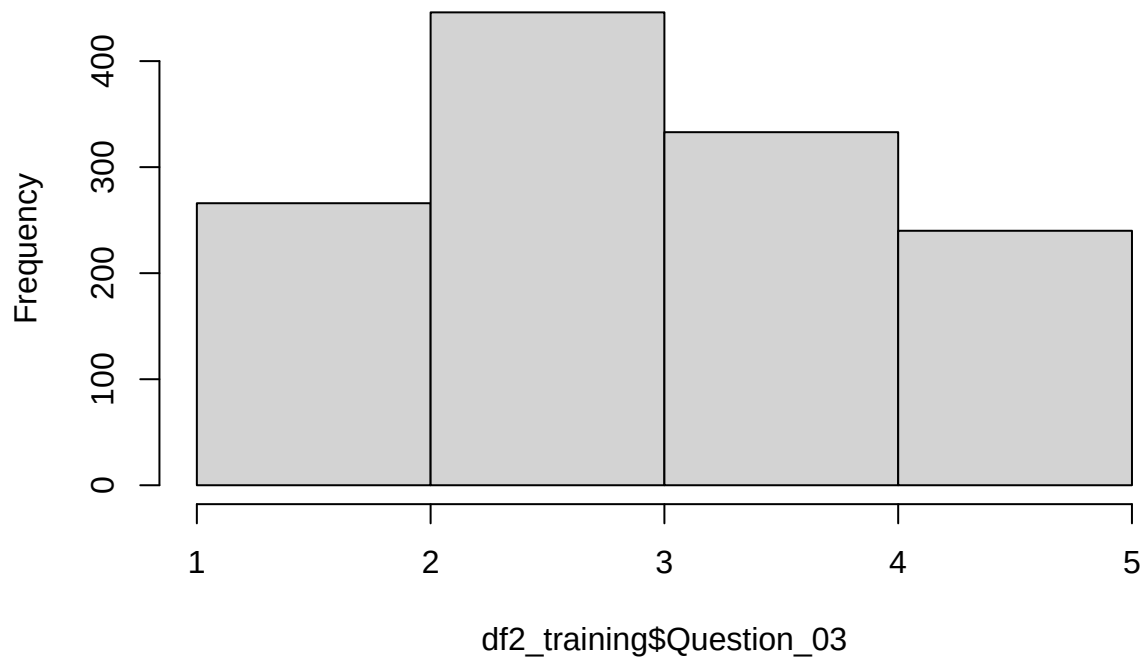
```
hist(df2_training$Question_02, breaks = 5)
```


Histogram of df2_training\$Question_02



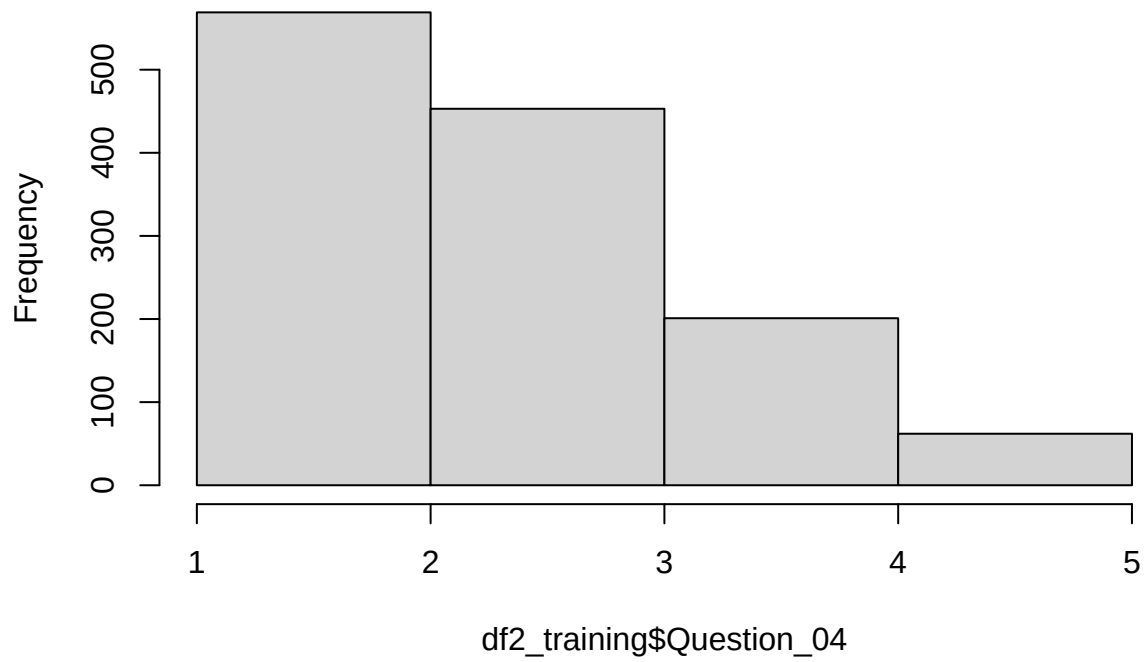
```
hist(df2_training$Question_03, breaks = 5)
```

Histogram of df2_training\$Question_03



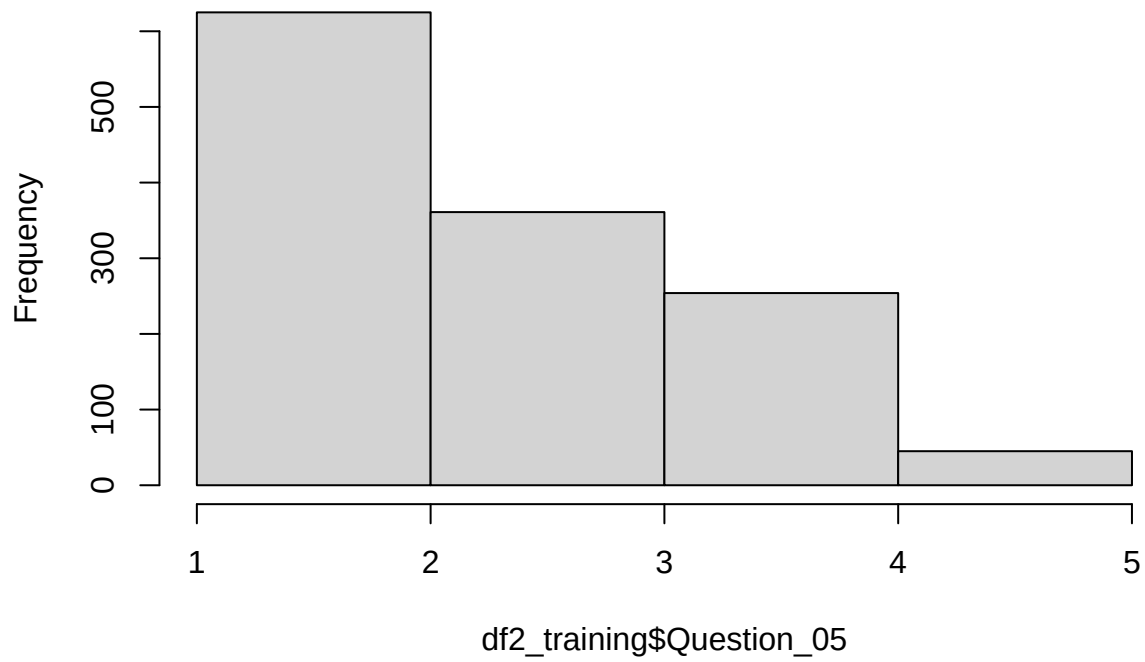
```
hist(df2_training$Question_04, breaks = 5)
```

Histogram of df2_training\$Question_04



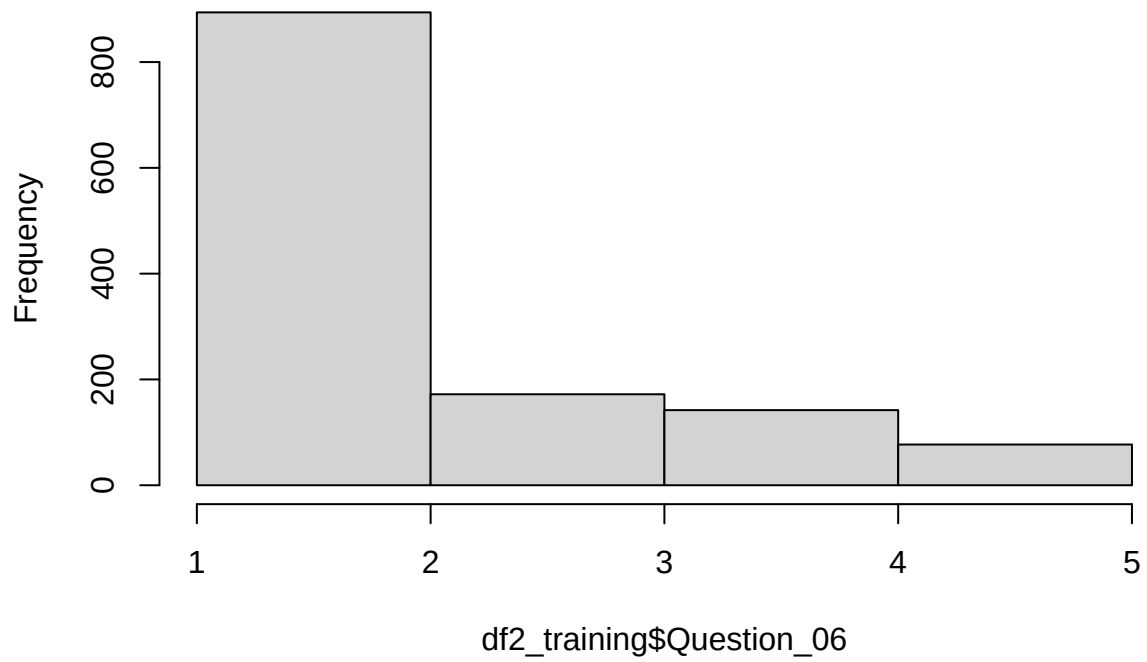
```
hist(df2_training$Question_05, breaks = 5)
```

Histogram of df2_training\$Question_05



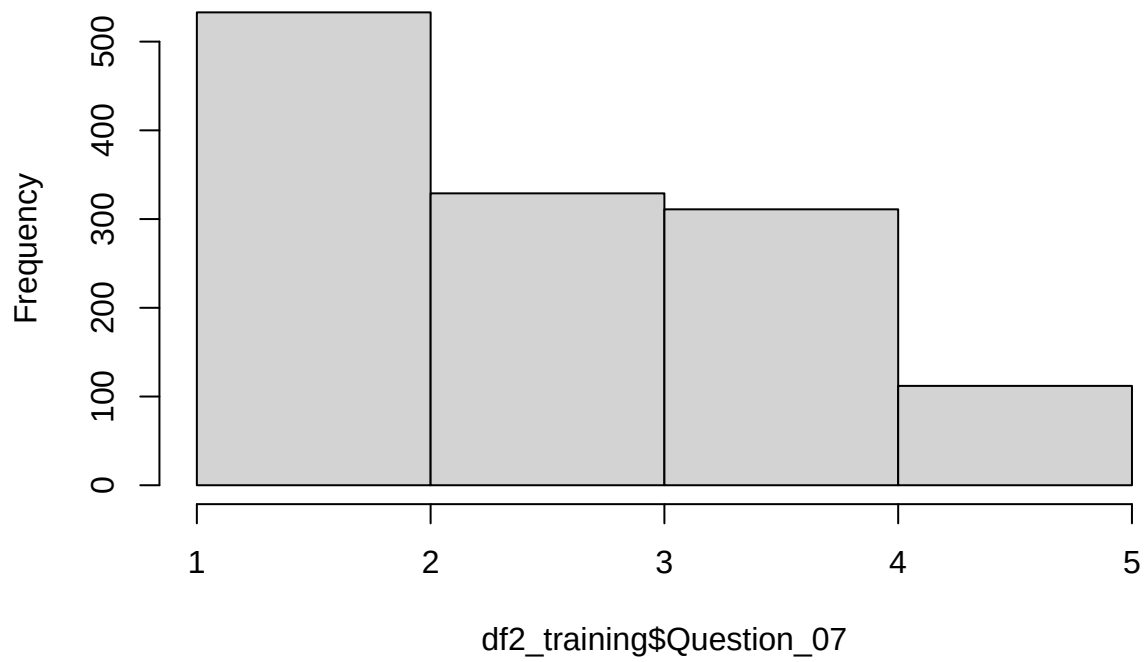
```
hist(df2_training$Question_06, breaks = 5)
```

Histogram of df2_training\$Question_06



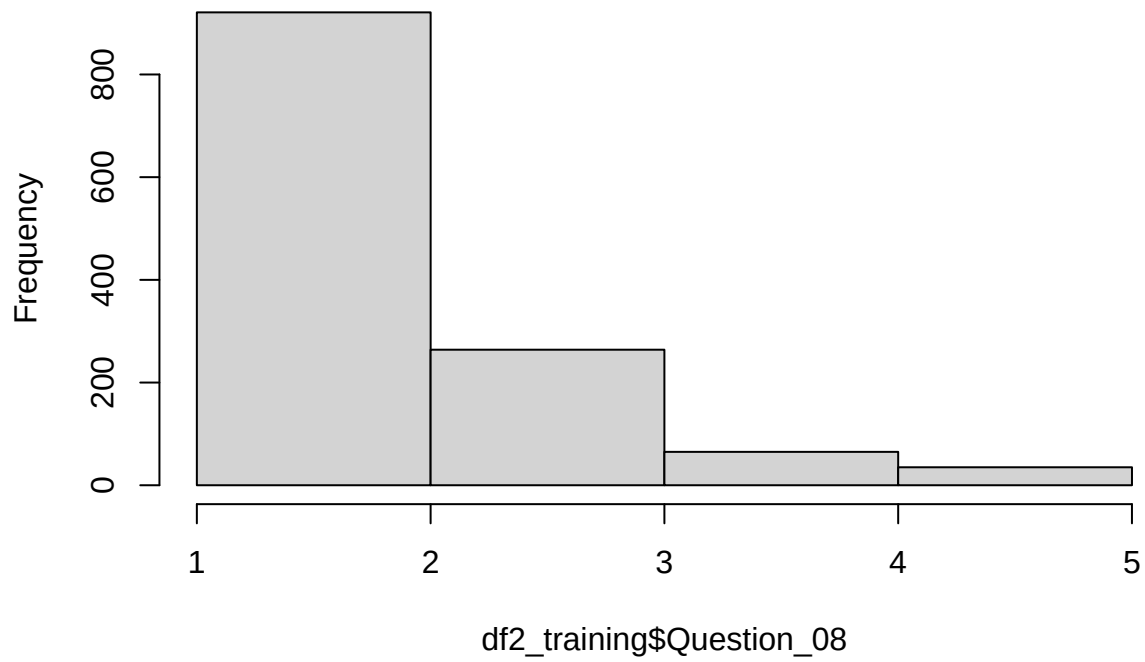
```
hist(df2_training$Question_07, breaks = 5)
```

Histogram of df2_training\$Question_07



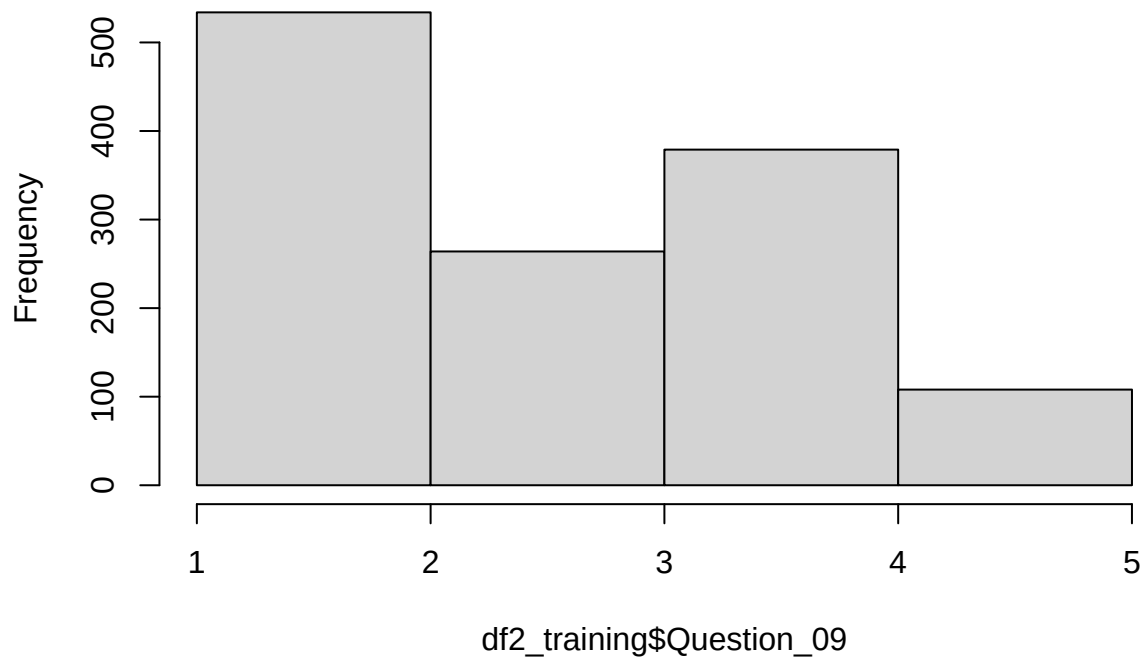
```
hist(df2_training$Question_08, breaks = 5)
```

Histogram of df2_training\$Question_08



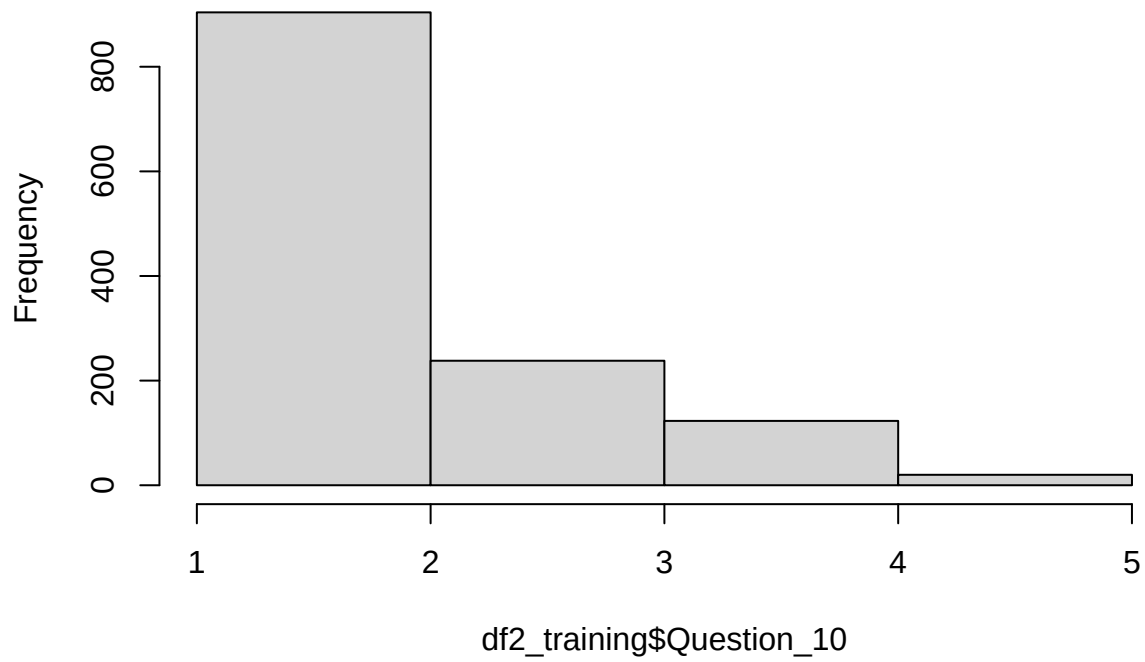
```
hist(df2_training$Question_09, breaks = 5)
```

Histogram of df2_training\$Question_09



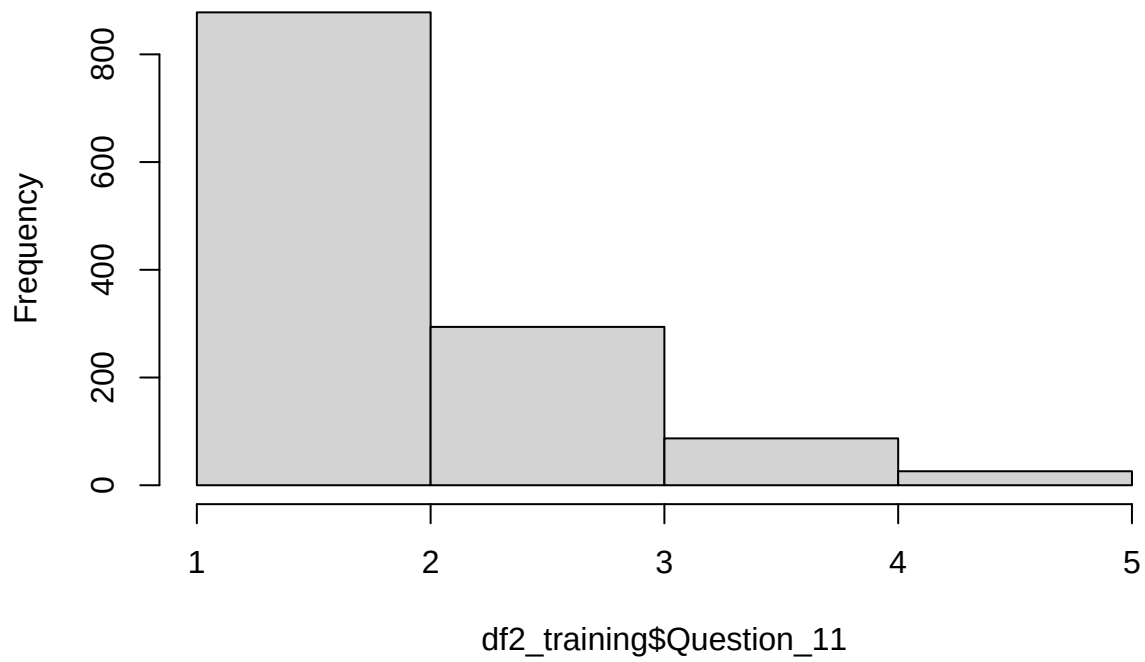
```
hist(df2_training$Question_10, breaks = 5)
```


Histogram of df2_training\$Question_10



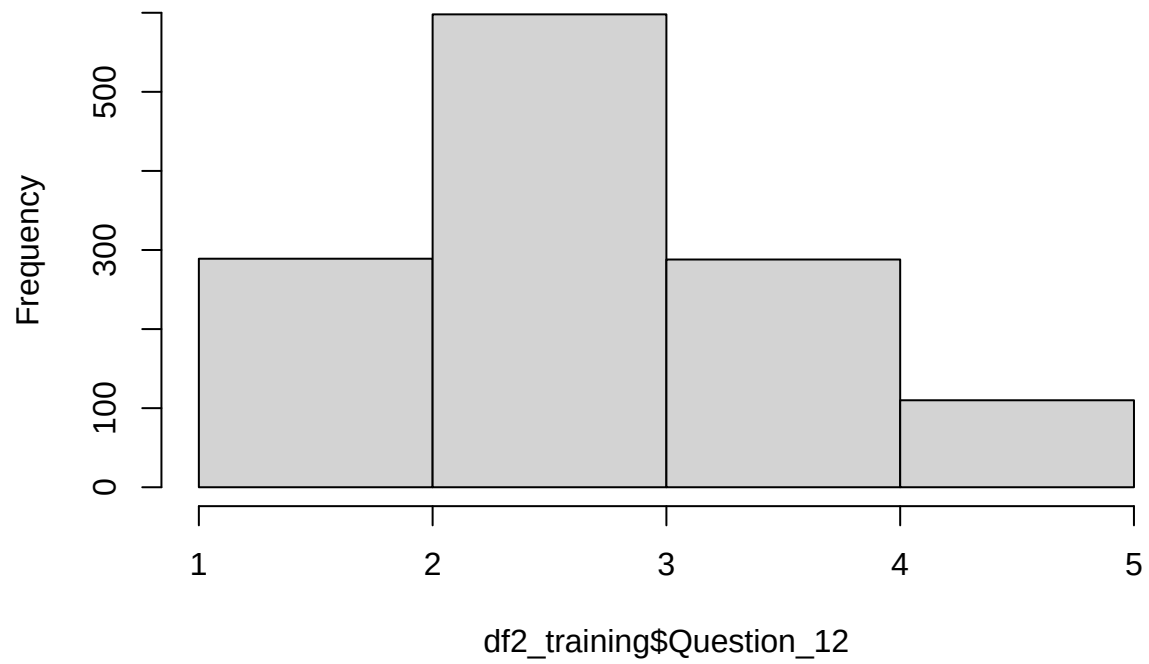
```
hist(df2_training$Question_11, breaks = 5)
```

Histogram of df2_training\$Question_11



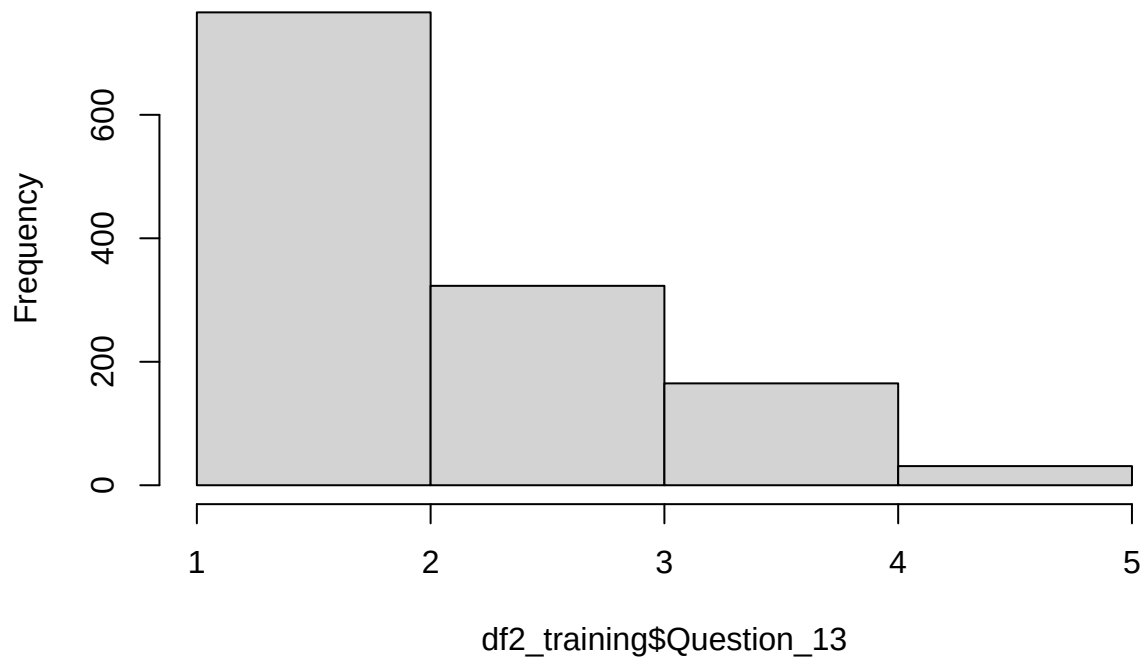
```
hist(df2_training$Question_12, breaks = 5)
```

Histogram of df2_training\$Question_12



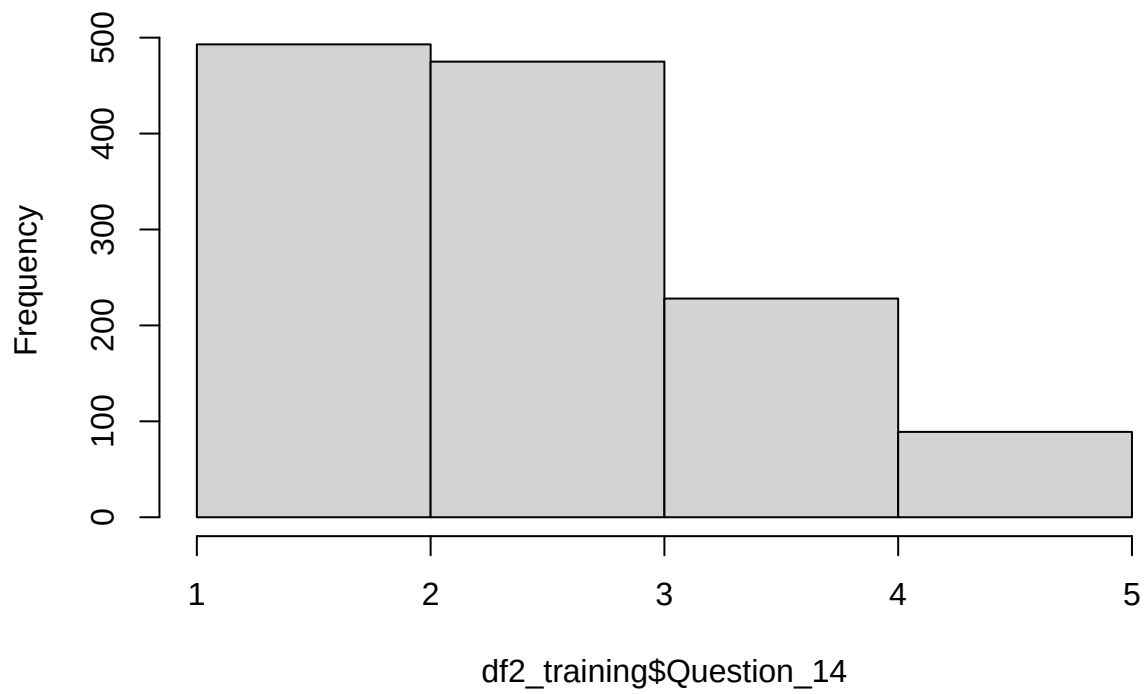
```
hist(df2_training$Question_13, breaks = 5)
```

Histogram of df2_training\$Question_13



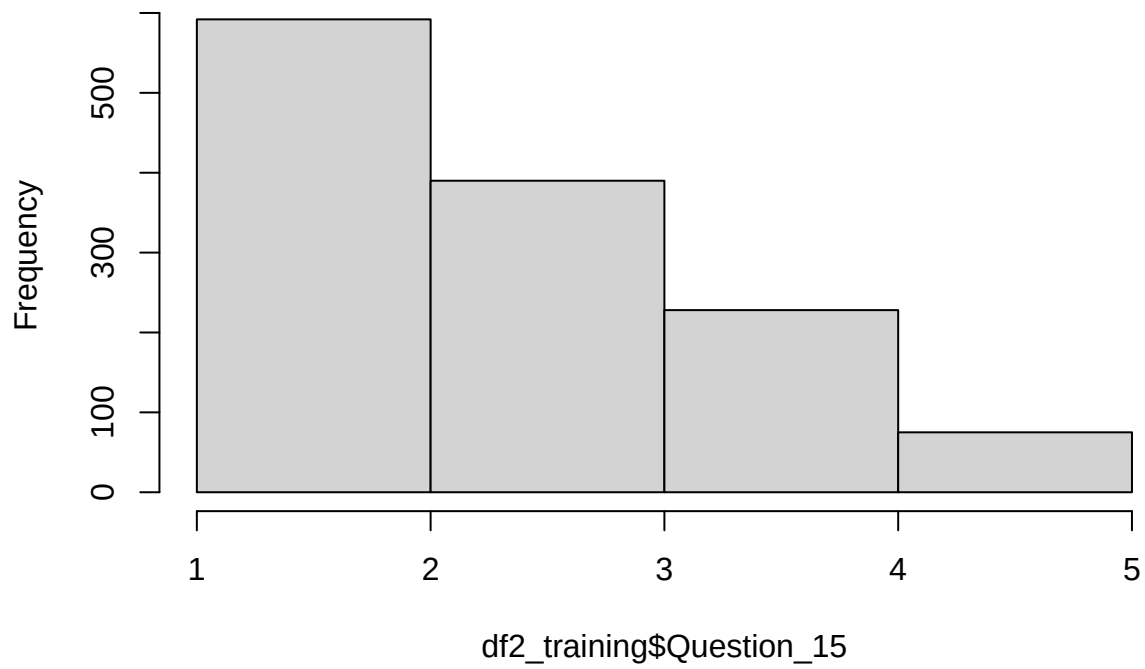
```
hist(df2_training$Question_14, breaks = 5)
```

Histogram of df2_training\$Question_14



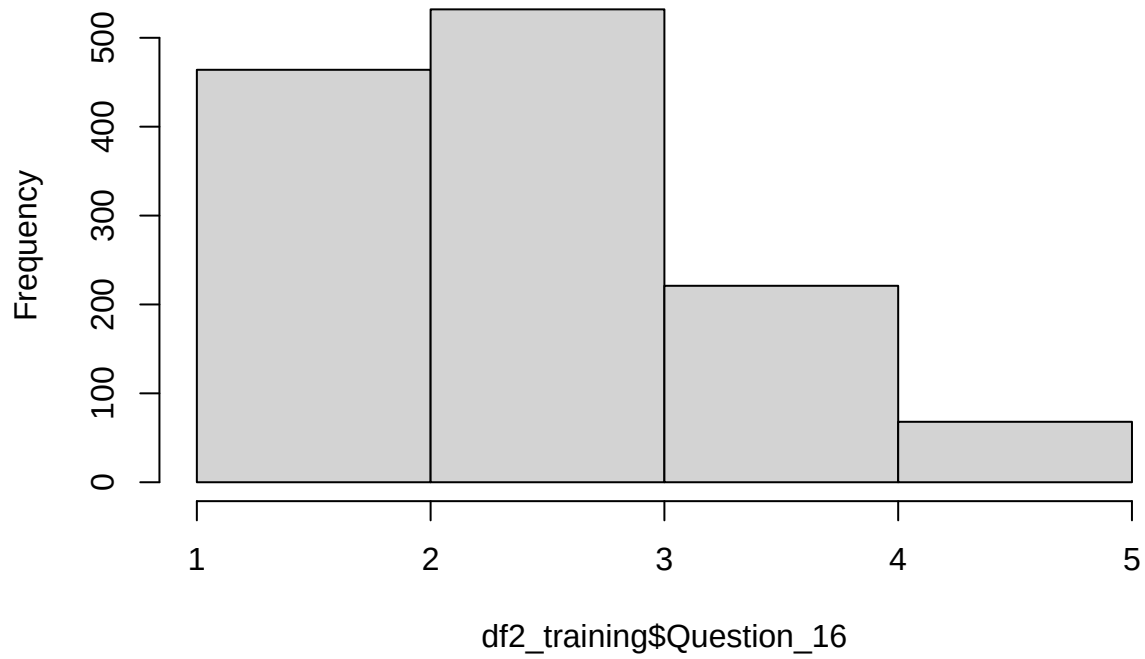
```
hist(df2_training$Question_15, breaks = 5)
```

Histogram of df2_training\$Question_15



```
hist(df2_training$Question_16, breaks = 5)
```

Histogram of df2_training\$Question_16

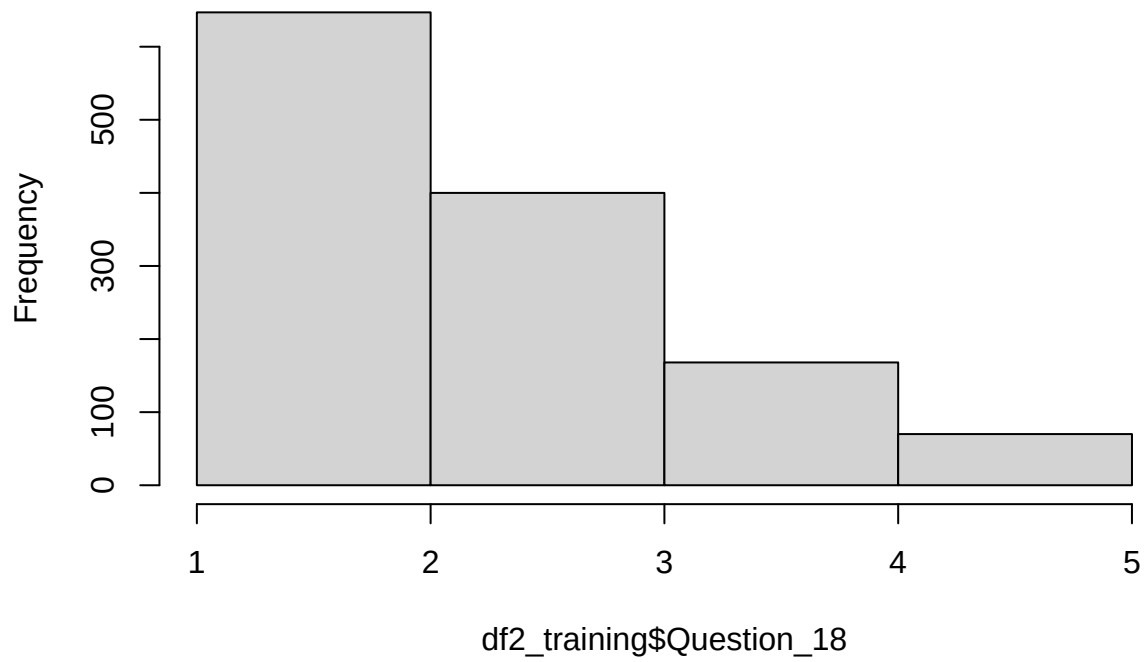


```
hist(df2_training$Question_17, breaks = 5)
```



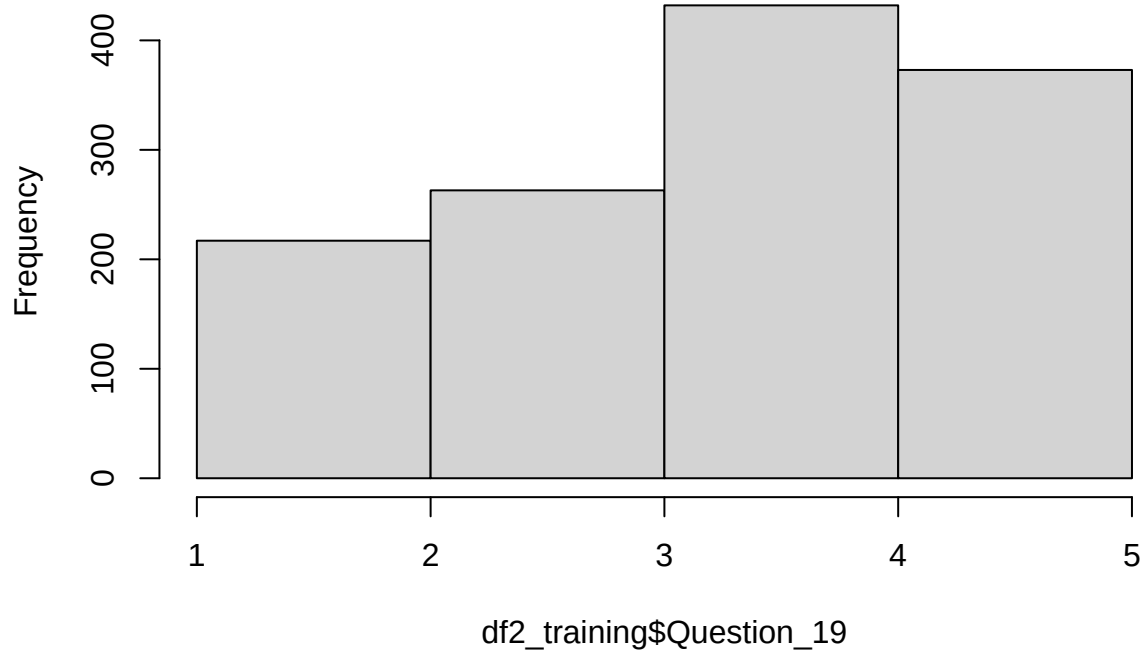
```
hist(df2_training$Question_18, breaks = 5)
```


Histogram of df2_training\$Question_18



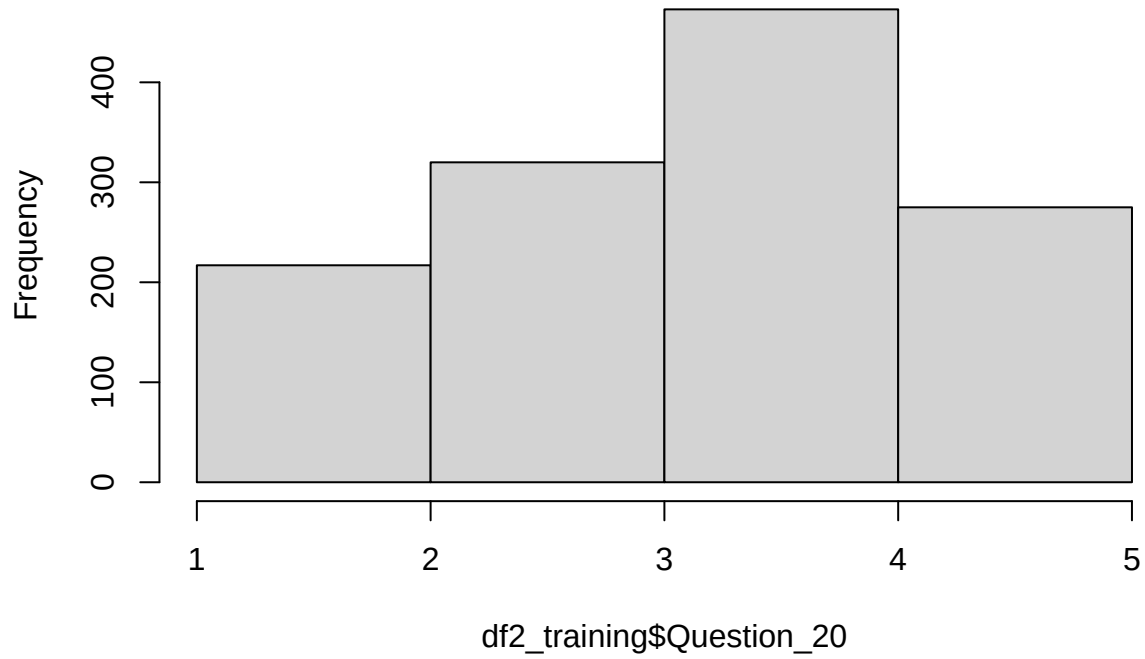
```
hist(df2_training$Question_19, breaks = 5)
```

Histogram of df2_training\$Question_19



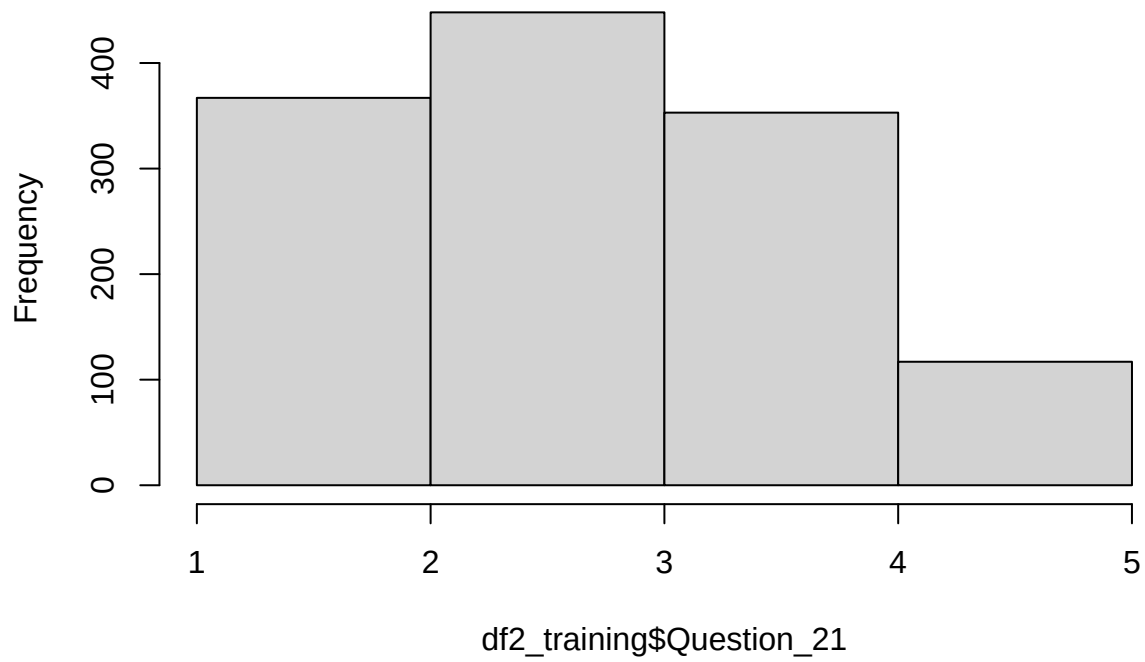
```
hist(df2_training$Question_20, breaks = 5)
```

Histogram of df2_training\$Question_20



```
hist(df2_training$Question_21, breaks = 5)
```

Histogram of df2_training\$Question_21

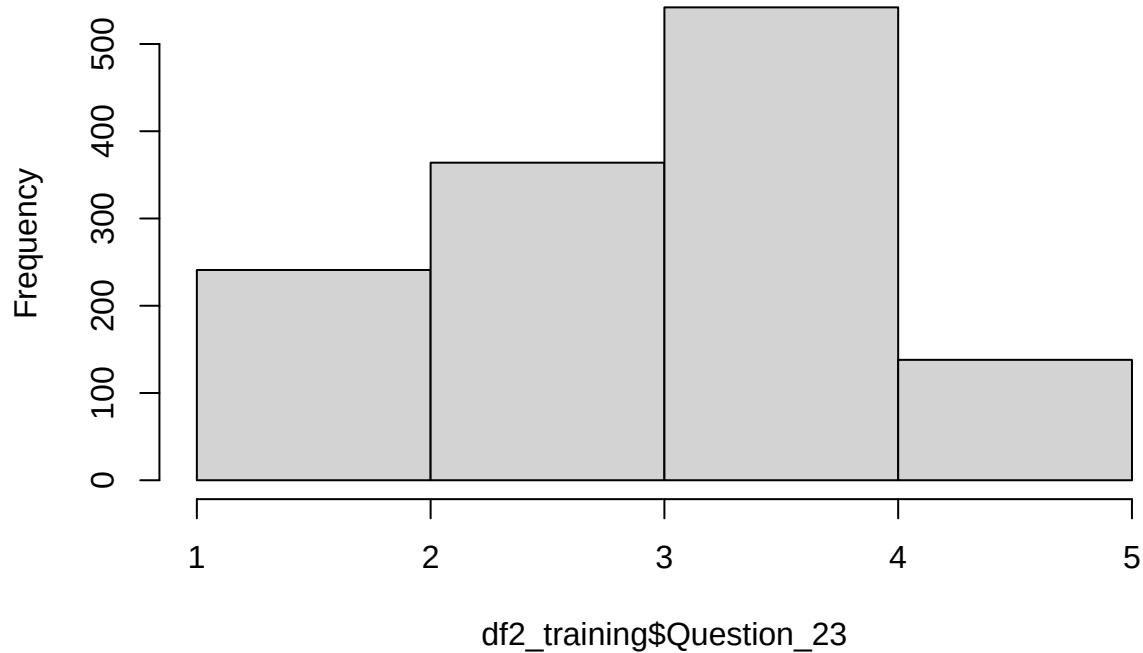


```
hist(df2_training$Question_22, breaks = 5)
```



```
hist(df2_training$Question_23, breaks = 5)
```

Histogram of df2_training\$Question_23



```
library(corr)

Cor_Mat <- df2_training %>%      #Now I will look at a correlation matrix of the question items
  correlate() %>%
  shave() %>% # Remove upper triangle
  fashion() # Print in nice format

## Correlation computed with
## * Method: 'pearson'
## * Missing treated using: 'pairwise.complete.obs'

print(Cor_Mat)

##           term   id Question_01 Question_02 Question_03 Question_04 Question_05
## 1           id
## 2 Question_01 .03
## 3 Question_02 .06          .09
## 4 Question_03 .11          .37          .33
## 5 Question_04 .03          .48          .12          .37
## 6 Question_05 .05          .42          .13          .30          .41
## 7 Question_06 .04          .26          .10          .26          .29          .26
## 8 Question_07 .05          .31          .17          .40          .41          .32
## 9 Question_08 .02          .34          .07          .27          .37          .27
## 10 Question_09 -.07        -.08        -.30        -.34        -.09        -.09
## 11 Question_10 -.02        .24          .09          .21          .23          .29
## 12 Question_11 .03          .34          .16          .36          .38          .28
## 13 Question_12 .03          .37          .19          .40          .44          .34
```

## 14	Question_13	.08	.37	.14	.32	.34	.31
## 15	Question_14	.09	.37	.20	.39	.37	.32
## 16	Question_15	.05	.25	.20	.33	.33	.26
## 17	Question_16	.04	.51	.19	.41	.41	.39
## 18	Question_17	.06	.39	.10	.33	.39	.32
## 19	Question_18	.02	.37	.18	.36	.39	.31
## 20	Question_19	.05	.20	.24	.36	.19	.16
## 21	Question_20	.06	.22	.18	.33	.25	.20
## 22	Question_21	.07	.36	.23	.41	.43	.33
## 23	Question_22	-.04	-.11	-.25	-.24	-.12	-.14
## 24	Question_23	-.10	-.03	-.12	-.18	-.02	-.04
## 25	FAC1_1	.06	.61	.21	.54	.62	.50
## 26	FAC2_1	.03	.23	.02	.17	.28	.26
## 27	FAC3_1	.01	.26	.01	.16	.26	.18
## 28	FAC4_1	-.10	.05	-.59	-.50	.04	-.00
## 29	FAC1_2	.06	.56	.24	.53	.56	.46
## 30	FAC2_2	-.09	.10	-.57	-.45	.09	.04
## 31	FAC3_2	.04	.27	.07	.24	.32	.30
## 32	FAC4_2	-.02	-.32	-.03	-.22	-.32	-.23
##	Question_06	Question_07	Question_08	Question_09	Question_10	Question_11	
## 1							
## 2							
## 3							
## 4							
## 5							
## 6							
## 7							
## 8	.51						
## 9	.24	.27					
## 10	-.14	-.15	.02				
## 11	.31	.29	.18	-.15			
## 12	.35	.34	.63	-.11	.28		
## 13	.34	.44	.28	-.18	.26	.37	
## 14	.47	.43	.33	-.16	.29	.40	
## 15	.40	.45	.31	-.14	.24	.33	
## 16	.34	.41	.30	-.20	.32	.36	
## 17	.27	.38	.32	-.19	.30	.36	
## 18	.31	.38	.58	-.03	.22	.57	
## 19	.50	.50	.32	-.16	.30	.38	
## 20	.22	.31	.17	-.27	.16	.25	
## 21	.11	.25	.20	-.15	.12	.28	
## 22	.27	.51	.29	-.15	.20	.36	
## 23	-.17	-.20	-.09	.28	-.12	-.19	
## 24	-.07	-.08	-.04	.21	-.04	-.08	
## 25	-.02	.39	.21	-.10	.15	.20	
## 26	.88	.63	.15	-.08	.41	.27	
## 27	.14	.12	.86	.08	.14	.78	
## 28	-.13	-.15	.00	.73	-.13	-.22	
## 29	-.11	.32	.12	-.14	.11	.12	
## 30	-.08	-.10	.02	.72	-.10	-.20	
## 31	.87	.66	.14	-.14	.43	.27	
## 32	-.16	-.17	-.88	-.07	-.17	-.80	
##	Question_12	Question_13	Question_14	Question_15	Question_16	Question_17	
## 1							

```

## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14      .51
## 15      .45      .45
## 16      .35      .35      .37
## 17      .41      .35      .43      .46
## 18      .33      .40      .36      .37      .40
## 19      .51      .52      .51      .35      .41      .39
## 20      .31      .27      .30      .25      .27      .17
## 21      .29      .21      .26      .23      .29      .21
## 22      .45      .36      .40      .30      .42      .34
## 23      -.19      -.17      -.23      -.19      -.20      -.15
## 24      -.05      -.03      -.09      -.06      -.11      -.08
## 25      .56      .28      .43      .29      .61      .29
## 26      .44      .63      .52      .39      .27      .28
## 27      .10      .22      .14      .25      .22      .71
## 28      -.15      -.13      -.16      -.26      -.16      -.04
## 29      .51      .21      .38      .24      .57      .20
## 30      -.09      -.08      -.10      -.22      -.11      -.01
## 31      .49      .65      .56      .42      .32      .29
## 32      -.17      -.27      -.20      -.29      -.29      -.74
##      Question_18 Question_19 Question_20 Question_21 Question_22 Question_23
## 1
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14
## 15
## 16
## 17
## 18
## 19
## 20      .28
## 21      .26      .26
## 22      .43      .28      .50

```



```

## 23      -.16      -.27      -.12      -.13
## 24      -.07      -.14      -.07      -.07      .25
## 25      .38      .28      .55      .70      -.01      .05
## 26      .68      .19      .01      .25      -.16      -.02
## 27      .14      .07      .09      .12      -.09      -.07
## 28      -.12      -.49      -.22      -.14      .60      .45
## 29      .31      .28      .55      .66      -.03      .03
## 30      -.06      -.46      -.19      -.08      .59      .45
## 31      .71      .24      .07      .32      -.19      -.04
## 32      -.19      -.10      -.15      -.19      .10      .07
##      FAC1_1 FAC2_1 FAC3_1 FAC4_1 FAC1_2 FAC2_2 FAC3_2 FAC4_2
## 1
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14
## 15
## 16
## 17
## 18
## 19
## 20
## 21
## 22
## 23
## 24
## 25
## 26      .03
## 27      .01      .02
## 28      .00      -.02      -.00
## 29      .99      -.07      -.08      -.06
## 30      .07      .03      -.00      1.00      .00
## 31      .12      .99      -.02      -.08      .03      -.02
## 32      -.10      -.05      -1.00      .01      -.01      .00      -.02

```

#That was very large so I will now run the Flatten Correlation Matrix Function

```

flattenCorrMatrix <- function(cormat, pmat, nmat) {
  ut <- upper.tri(cormat)
  data.frame(
    row = rownames(cormat)[row(cormat)[ut]],
    column = rownames(cormat)[col(cormat)[ut]],
    cor = (cormat)[ut],
    p = pmat[ut],
    n = nmat[ut]
  )
}

```

```

)
}

#install.packages("Hmisc", dependencies = TRUE)
library(Hmisc)
df2_training_MAT <- as.matrix(df2_training) # I am changing my df into a Matrix

library(Hmisc)
#install.packages("checkmate", dependencies = TRUE)
library(checkmate) # This is to get the correlation matrix to feed into our `flattenCorrMatrix` fun
res <- rcorr(df2_training_MAT)
print(res)

```

```

##          id Question_01 Question_02 Question_03 Question_04 Question_05
## id      1.00         0.03         0.06         0.11         0.03         0.05
## Question_01 0.03         1.00         0.09         0.37         0.48         0.42
## Question_02 0.06         0.09         1.00         0.33         0.12         0.13
## Question_03 0.11         0.37         0.33         1.00         0.37         0.30
## Question_04 0.03         0.48         0.12         0.37         1.00         0.41
## Question_05 0.05         0.42         0.13         0.30         0.41         1.00
## Question_06 0.04         0.26         0.10         0.26         0.29         0.26
## Question_07 0.05         0.31         0.17         0.40         0.41         0.32
## Question_08 0.02         0.34         0.07         0.27         0.37         0.27
## Question_09 -0.07        -0.08        -0.30        -0.34        -0.09        -0.09
## Question_10 -0.02         0.24         0.09         0.21         0.23         0.29
## Question_11 0.03         0.34         0.16         0.36         0.38         0.28
## Question_12 0.03         0.37         0.19         0.40         0.44         0.34
## Question_13 0.08         0.37         0.14         0.32         0.34         0.31
## Question_14 0.09         0.37         0.20         0.39         0.37         0.32
## Question_15 0.05         0.25         0.20         0.33         0.33         0.26
## Question_16 0.04         0.51         0.19         0.41         0.41         0.39
## Question_17 0.06         0.39         0.10         0.33         0.39         0.32
## Question_18 0.02         0.37         0.18         0.36         0.39         0.31
## Question_19 0.05         0.20         0.24         0.36         0.19         0.16
## Question_20 0.06         0.22         0.18         0.33         0.25         0.20
## Question_21 0.07         0.36         0.23         0.41         0.43         0.33
## Question_22 -0.04        -0.11        -0.25        -0.24        -0.12        -0.14
## Question_23 -0.10        -0.03        -0.12        -0.18        -0.02        -0.04
## FAC1_1      0.06         0.61         0.21         0.54         0.62         0.50
## FAC2_1      0.03         0.23         0.02         0.17         0.28         0.26
## FAC3_1      0.01         0.26         0.01         0.16         0.26         0.18
## FAC4_1     -0.10         0.05        -0.59        -0.50         0.04         0.00
## FAC1_2      0.06         0.56         0.24         0.53         0.56         0.46
## FAC2_2     -0.09         0.10        -0.57        -0.45         0.09         0.04
## FAC3_2      0.04         0.27         0.07         0.24         0.32         0.30
## FAC4_2     -0.02        -0.32        -0.03        -0.22        -0.32        -0.23
##          Question_06 Question_07 Question_08 Question_09 Question_10
## id          0.04         0.05         0.02        -0.07        -0.02
## Question_01  0.26         0.31         0.34        -0.08         0.24
## Question_02  0.10         0.17         0.07        -0.30         0.09
## Question_03  0.26         0.40         0.27        -0.34         0.21
## Question_04  0.29         0.41         0.37        -0.09         0.23
## Question_05  0.26         0.32         0.27        -0.09         0.29
## Question_06  1.00         0.51         0.24        -0.14         0.31

```

## Question_07	0.51	1.00	0.27	-0.15	0.29
## Question_08	0.24	0.27	1.00	0.02	0.18
## Question_09	-0.14	-0.15	0.02	1.00	-0.15
## Question_10	0.31	0.29	0.18	-0.15	1.00
## Question_11	0.35	0.34	0.63	-0.11	0.28
## Question_12	0.34	0.44	0.28	-0.18	0.26
## Question_13	0.47	0.43	0.33	-0.16	0.29
## Question_14	0.40	0.45	0.31	-0.14	0.24
## Question_15	0.34	0.41	0.30	-0.20	0.32
## Question_16	0.27	0.38	0.32	-0.19	0.30
## Question_17	0.31	0.38	0.58	-0.03	0.22
## Question_18	0.50	0.50	0.32	-0.16	0.30
## Question_19	0.22	0.31	0.17	-0.27	0.16
## Question_20	0.11	0.25	0.20	-0.15	0.12
## Question_21	0.27	0.51	0.29	-0.15	0.20
## Question_22	-0.17	-0.20	-0.09	0.28	-0.12
## Question_23	-0.07	-0.08	-0.04	0.21	-0.04
## FAC1_1	-0.02	0.39	0.21	-0.10	0.15
## FAC2_1	0.88	0.63	0.15	-0.08	0.41
## FAC3_1	0.14	0.12	0.86	0.08	0.14
## FAC4_1	-0.13	-0.15	0.00	0.73	-0.13
## FAC1_2	-0.11	0.32	0.12	-0.14	0.11
## FAC2_2	-0.08	-0.10	0.02	0.72	-0.10
## FAC3_2	0.87	0.66	0.14	-0.14	0.43
## FAC4_2	-0.16	-0.17	-0.88	-0.07	-0.17
##	Question_11	Question_12	Question_13	Question_14	Question_15
## id	0.03	0.03	0.08	0.09	0.05
## Question_01	0.34	0.37	0.37	0.37	0.25
## Question_02	0.16	0.19	0.14	0.20	0.20
## Question_03	0.36	0.40	0.32	0.39	0.33
## Question_04	0.38	0.44	0.34	0.37	0.33
## Question_05	0.28	0.34	0.31	0.32	0.26
## Question_06	0.35	0.34	0.47	0.40	0.34
## Question_07	0.34	0.44	0.43	0.45	0.41
## Question_08	0.63	0.28	0.33	0.31	0.30
## Question_09	-0.11	-0.18	-0.16	-0.14	-0.20
## Question_10	0.28	0.26	0.29	0.24	0.32
## Question_11	1.00	0.37	0.40	0.33	0.36
## Question_12	0.37	1.00	0.51	0.45	0.35
## Question_13	0.40	0.51	1.00	0.45	0.35
## Question_14	0.33	0.45	0.45	1.00	0.37
## Question_15	0.36	0.35	0.35	0.37	1.00
## Question_16	0.36	0.41	0.35	0.43	0.46
## Question_17	0.57	0.33	0.40	0.36	0.37
## Question_18	0.38	0.51	0.52	0.51	0.35
## Question_19	0.25	0.31	0.27	0.30	0.25
## Question_20	0.28	0.29	0.21	0.26	0.23
## Question_21	0.36	0.45	0.36	0.40	0.30
## Question_22	-0.19	-0.19	-0.17	-0.23	-0.19
## Question_23	-0.08	-0.05	-0.03	-0.09	-0.06
## FAC1_1	0.20	0.56	0.28	0.43	0.29
## FAC2_1	0.27	0.44	0.63	0.52	0.39
## FAC3_1	0.78	0.10	0.22	0.14	0.25
## FAC4_1	-0.22	-0.15	-0.13	-0.16	-0.26

##	FAC1_2	0.12	0.51	0.21	0.38	0.24		
##	FAC2_2	-0.20	-0.09	-0.08	-0.10	-0.22		
##	FAC3_2	0.27	0.49	0.65	0.56	0.42		
##	FAC4_2	-0.80	-0.17	-0.27	-0.20	-0.29		
##		Question_16	Question_17	Question_18	Question_19	Question_20		
##	id	0.04	0.06	0.02	0.05	0.06		
##	Question_01	0.51	0.39	0.37	0.20	0.22		
##	Question_02	0.19	0.10	0.18	0.24	0.18		
##	Question_03	0.41	0.33	0.36	0.36	0.33		
##	Question_04	0.41	0.39	0.39	0.19	0.25		
##	Question_05	0.39	0.32	0.31	0.16	0.20		
##	Question_06	0.27	0.31	0.50	0.22	0.11		
##	Question_07	0.38	0.38	0.50	0.31	0.25		
##	Question_08	0.32	0.58	0.32	0.17	0.20		
##	Question_09	-0.19	-0.03	-0.16	-0.27	-0.15		
##	Question_10	0.30	0.22	0.30	0.16	0.12		
##	Question_11	0.36	0.57	0.38	0.25	0.28		
##	Question_12	0.41	0.33	0.51	0.31	0.29		
##	Question_13	0.35	0.40	0.52	0.27	0.21		
##	Question_14	0.43	0.36	0.51	0.30	0.26		
##	Question_15	0.46	0.37	0.35	0.25	0.23		
##	Question_16	1.00	0.40	0.41	0.27	0.29		
##	Question_17	0.40	1.00	0.39	0.17	0.21		
##	Question_18	0.41	0.39	1.00	0.28	0.26		
##	Question_19	0.27	0.17	0.28	1.00	0.26		
##	Question_20	0.29	0.21	0.26	0.26	1.00		
##	Question_21	0.42	0.34	0.43	0.28	0.50		
##	Question_22	-0.20	-0.15	-0.16	-0.27	-0.12		
##	Question_23	-0.11	-0.08	-0.07	-0.14	-0.07		
##	FAC1_1	0.61	0.29	0.38	0.28	0.55		
##	FAC2_1	0.27	0.28	0.68	0.19	0.01		
##	FAC3_1	0.22	0.71	0.14	0.07	0.09		
##	FAC4_1	-0.16	-0.04	-0.12	-0.49	-0.22		
##	FAC1_2	0.57	0.20	0.31	0.28	0.55		
##	FAC2_2	-0.11	-0.01	-0.06	-0.46	-0.19		
##	FAC3_2	0.32	0.29	0.71	0.24	0.07		
##	FAC4_2	-0.29	-0.74	-0.19	-0.10	-0.15		
##		Question_21	Question_22	Question_23	FAC1_1	FAC2_1	FAC3_1	FAC4_1
##	id	0.07	-0.04	-0.10	0.06	0.03	0.01	-0.10
##	Question_01	0.36	-0.11	-0.03	0.61	0.23	0.26	0.05
##	Question_02	0.23	-0.25	-0.12	0.21	0.02	0.01	-0.59
##	Question_03	0.41	-0.24	-0.18	0.54	0.17	0.16	-0.50
##	Question_04	0.43	-0.12	-0.02	0.62	0.28	0.26	0.04
##	Question_05	0.33	-0.14	-0.04	0.50	0.26	0.18	0.00
##	Question_06	0.27	-0.17	-0.07	-0.02	0.88	0.14	-0.13
##	Question_07	0.51	-0.20	-0.08	0.39	0.63	0.12	-0.15
##	Question_08	0.29	-0.09	-0.04	0.21	0.15	0.86	0.00
##	Question_09	-0.15	0.28	0.21	-0.10	-0.08	0.08	0.73
##	Question_10	0.20	-0.12	-0.04	0.15	0.41	0.14	-0.13
##	Question_11	0.36	-0.19	-0.08	0.20	0.27	0.78	-0.22
##	Question_12	0.45	-0.19	-0.05	0.56	0.44	0.10	-0.15
##	Question_13	0.36	-0.17	-0.03	0.28	0.63	0.22	-0.13
##	Question_14	0.40	-0.23	-0.09	0.43	0.52	0.14	-0.16
##	Question_15	0.30	-0.19	-0.06	0.29	0.39	0.25	-0.26

```

## Question_16      0.42      -0.20      -0.11      0.61      0.27      0.22      -0.16
## Question_17      0.34      -0.15      -0.08      0.29      0.28      0.71      -0.04
## Question_18      0.43      -0.16      -0.07      0.38      0.68      0.14      -0.12
## Question_19      0.28      -0.27      -0.14      0.28      0.19      0.07      -0.49
## Question_20      0.50      -0.12      -0.07      0.55      0.01      0.09      -0.22
## Question_21      1.00      -0.13      -0.07      0.70      0.25      0.12      -0.14
## Question_22      -0.13      1.00      0.25      -0.01      -0.16      -0.09      0.60
## Question_23      -0.07      0.25      1.00      0.05      -0.02      -0.07      0.45
## FAC1_1           0.70      -0.01      0.05      1.00      0.03      0.01      0.00
## FAC2_1           0.25      -0.16      -0.02      0.03      1.00      0.02      -0.02
## FAC3_1           0.12      -0.09      -0.07      0.01      0.02      1.00      0.00
## FAC4_1          -0.14      0.60      0.45      0.00      -0.02      0.00      1.00
## FAC1_2           0.66      -0.03      0.03      0.99      -0.07      -0.08      -0.06
## FAC2_2          -0.08      0.59      0.45      0.07      0.03      0.00      1.00
## FAC3_2           0.32      -0.19      -0.04      0.12      0.99      -0.02      -0.08
## FAC4_2          -0.19      0.10      0.07      -0.10      -0.05      -1.00      0.01
##
##      FAC1_2 FAC2_2 FAC3_2 FAC4_2
## id      0.06 -0.09  0.04 -0.02
## Question_01 0.56  0.10  0.27 -0.32
## Question_02 0.24 -0.57  0.07 -0.03
## Question_03 0.53 -0.45  0.24 -0.22
## Question_04 0.56  0.09  0.32 -0.32
## Question_05 0.46  0.04  0.30 -0.23
## Question_06 -0.11 -0.08  0.87 -0.16
## Question_07 0.32 -0.10  0.66 -0.17
## Question_08 0.12  0.02  0.14 -0.88
## Question_09 -0.14  0.72 -0.14 -0.07
## Question_10 0.11 -0.10  0.43 -0.17
## Question_11 0.12 -0.20  0.27 -0.80
## Question_12 0.51 -0.09  0.49 -0.17
## Question_13 0.21 -0.08  0.65 -0.27
## Question_14 0.38 -0.10  0.56 -0.20
## Question_15 0.24 -0.22  0.42 -0.29
## Question_16 0.57 -0.11  0.32 -0.29
## Question_17 0.20 -0.01  0.29 -0.74
## Question_18 0.31 -0.06  0.71 -0.19
## Question_19 0.28 -0.46  0.24 -0.10
## Question_20 0.55 -0.19  0.07 -0.15
## Question_21 0.66 -0.08  0.32 -0.19
## Question_22 -0.03  0.59 -0.19  0.10
## Question_23 0.03  0.45 -0.04  0.07
## FAC1_1      0.99  0.07  0.12 -0.10
## FAC2_1     -0.07  0.03  0.99 -0.05
## FAC3_1     -0.08  0.00 -0.02 -1.00
## FAC4_1     -0.06  1.00 -0.08  0.01
## FAC1_2      1.00  0.00  0.03 -0.01
## FAC2_2      0.00  1.00 -0.02  0.00
## FAC3_2      0.03 -0.02  1.00 -0.02
## FAC4_2     -0.01  0.00 -0.02  1.00
##
## n= 1285
##
##
## P

```

##	id	Question_01	Question_02	Question_03	Question_04	Question_05
## id		0.2248	0.0335	0.0001	0.2572	0.0641
## Question_01	0.2248		0.0014	0.0000	0.0000	0.0000
## Question_02	0.0335	0.0014		0.0000	0.0000	0.0000
## Question_03	0.0001	0.0000	0.0000		0.0000	0.0000
## Question_04	0.2572	0.0000	0.0000	0.0000		0.0000
## Question_05	0.0641	0.0000	0.0000	0.0000	0.0000	
## Question_06	0.1490	0.0000	0.0006	0.0000	0.0000	0.0000
## Question_07	0.0590	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_08	0.5604	0.0000	0.0105	0.0000	0.0000	0.0000
## Question_09	0.0126	0.0030	0.0000	0.0000	0.0007	0.0007
## Question_10	0.3896	0.0000	0.0012	0.0000	0.0000	0.0000
## Question_11	0.2556	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_12	0.2995	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_13	0.0043	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_14	0.0013	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_15	0.0506	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_16	0.1547	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_17	0.0226	0.0000	0.0002	0.0000	0.0000	0.0000
## Question_18	0.3724	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_19	0.0897	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_20	0.0220	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_21	0.0084	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_22	0.1137	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_23	0.0002	0.2838	0.0000	0.0000	0.4852	0.1863
## FAC1_1	0.0448	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC2_1	0.3257	0.0000	0.5048	0.0000	0.0000	0.0000
## FAC3_1	0.6330	0.0000	0.8489	0.0000	0.0000	0.0000
## FAC4_1	0.0005	0.0562	0.0000	0.0000	0.1883	0.9405
## FAC1_2	0.0379	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC2_2	0.0010	0.0002	0.0000	0.0000	0.0011	0.1163
## FAC3_2	0.1747	0.0000	0.0079	0.0000	0.0000	0.0000
## FAC4_2	0.4811	0.0000	0.3236	0.0000	0.0000	0.0000
##	Question_06	Question_07	Question_08	Question_09	Question_10	
## id	0.1490	0.0590	0.5604	0.0126	0.3896	
## Question_01	0.0000	0.0000	0.0000	0.0030	0.0000	
## Question_02	0.0006	0.0000	0.0105	0.0000	0.0012	
## Question_03	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_04	0.0000	0.0000	0.0000	0.0007	0.0000	
## Question_05	0.0000	0.0000	0.0000	0.0007	0.0000	
## Question_06		0.0000	0.0000	0.0000	0.0000	
## Question_07	0.0000		0.0000	0.0000	0.0000	
## Question_08	0.0000	0.0000		0.5770	0.0000	
## Question_09	0.0000	0.0000	0.5770		0.0000	
## Question_10	0.0000	0.0000	0.0000	0.0000		
## Question_11	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_12	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_13	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_14	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_15	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_16	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_17	0.0000	0.0000	0.0000	0.2964	0.0000	
## Question_18	0.0000	0.0000	0.0000	0.0000	0.0000	
## Question_19	0.0000	0.0000	0.0000	0.0000	0.0000	

## Question_20	0.0001	0.0000	0.0000	0.0000	0.0000
## Question_21	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_22	0.0000	0.0000	0.0016	0.0000	0.0000
## Question_23	0.0179	0.0058	0.1747	0.0000	0.1629
## FAC1_1	0.5499	0.0000	0.0000	0.0005	0.0000
## FAC2_1	0.0000	0.0000	0.0000	0.0029	0.0000
## FAC3_1	0.0000	0.0000	0.0000	0.0029	0.0000
## FAC4_1	0.0000	0.0000	0.9460	0.0000	0.0000
## FAC1_2	0.0001	0.0000	0.0000	0.0000	0.0001
## FAC2_2	0.0032	0.0006	0.4671	0.0000	0.0002
## FAC3_2	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC4_2	0.0000	0.0000	0.0000	0.0145	0.0000
##	Question_11	Question_12	Question_13	Question_14	Question_15
## id	0.2556	0.2995	0.0043	0.0013	0.0506
## Question_01	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_02	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_03	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_04	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_05	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_06	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_07	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_08	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_09	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_10	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_11		0.0000	0.0000	0.0000	0.0000
## Question_12	0.0000		0.0000	0.0000	0.0000
## Question_13	0.0000	0.0000		0.0000	0.0000
## Question_14	0.0000	0.0000	0.0000		0.0000
## Question_15	0.0000	0.0000	0.0000	0.0000	
## Question_16	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_17	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_18	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_19	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_20	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_21	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_22	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_23	0.0066	0.0987	0.2552	0.0018	0.0387
## FAC1_1	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC2_1	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC3_1	0.0000	0.0003	0.0000	0.0000	0.0000
## FAC4_1	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC1_2	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC2_2	0.0000	0.0015	0.0043	0.0002	0.0000
## FAC3_2	0.0000	0.0000	0.0000	0.0000	0.0000
## FAC4_2	0.0000	0.0000	0.0000	0.0000	0.0000
##	Question_16	Question_17	Question_18	Question_19	Question_20
## id	0.1547	0.0226	0.3724	0.0897	0.0220
## Question_01	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_02	0.0000	0.0002	0.0000	0.0000	0.0000
## Question_03	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_04	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_05	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_06	0.0000	0.0000	0.0000	0.0000	0.0001
## Question_07	0.0000	0.0000	0.0000	0.0000	0.0000

## Question_08	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_09	0.0000	0.2964	0.0000	0.0000	0.0000		
## Question_10	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_11	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_12	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_13	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_14	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_15	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_16		0.0000	0.0000	0.0000	0.0000		
## Question_17	0.0000		0.0000	0.0000	0.0000		
## Question_18	0.0000	0.0000		0.0000	0.0000		
## Question_19	0.0000	0.0000	0.0000		0.0000		
## Question_20	0.0000	0.0000	0.0000	0.0000			
## Question_21	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_22	0.0000	0.0000	0.0000	0.0000	0.0000		
## Question_23	0.0001	0.0037	0.0177	0.0000	0.0172		
## FAC1_1	0.0000	0.0000	0.0000	0.0000	0.0000		
## FAC2_1	0.0000	0.0000	0.0000	0.0000	0.7655		
## FAC3_1	0.0000	0.0000	0.0000	0.0146	0.0007		
## FAC4_1	0.0000	0.1366	0.0000	0.0000	0.0000		
## FAC1_2	0.0000	0.0000	0.0000	0.0000	0.0000		
## FAC2_2	0.0000	0.7104	0.0461	0.0000	0.0000		
## FAC3_2	0.0000	0.0000	0.0000	0.0000	0.0122		
## FAC4_2	0.0000	0.0000	0.0000	0.0003	0.0000		
##	Question_21	Question_22	Question_23	FAC1_1	FAC2_1	FAC3_1	FAC4_1
## id	0.0084	0.1137	0.0002	0.0448	0.3257	0.6330	0.0005
## Question_01	0.0000	0.0000	0.2838	0.0000	0.0000	0.0000	0.0562
## Question_02	0.0000	0.0000	0.0000	0.0000	0.5048	0.8489	0.0000
## Question_03	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
## Question_04	0.0000	0.0000	0.4852	0.0000	0.0000	0.0000	0.1883
## Question_05	0.0000	0.0000	0.1863	0.0000	0.0000	0.0000	0.9405
## Question_06	0.0000	0.0000	0.0179	0.5499	0.0000	0.0000	0.0000
## Question_07	0.0000	0.0000	0.0058	0.0000	0.0000	0.0000	0.0000
## Question_08	0.0000	0.0016	0.1747	0.0000	0.0000	0.0000	0.9460
## Question_09	0.0000	0.0000	0.0000	0.0005	0.0029	0.0029	0.0000
## Question_10	0.0000	0.0000	0.1629	0.0000	0.0000	0.0000	0.0000
## Question_11	0.0000	0.0000	0.0066	0.0000	0.0000	0.0000	0.0000
## Question_12	0.0000	0.0000	0.0987	0.0000	0.0000	0.0003	0.0000
## Question_13	0.0000	0.0000	0.2552	0.0000	0.0000	0.0000	0.0000
## Question_14	0.0000	0.0000	0.0018	0.0000	0.0000	0.0000	0.0000
## Question_15	0.0000	0.0000	0.0387	0.0000	0.0000	0.0000	0.0000
## Question_16	0.0000	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000
## Question_17	0.0000	0.0000	0.0037	0.0000	0.0000	0.0000	0.1366
## Question_18	0.0000	0.0000	0.0177	0.0000	0.0000	0.0000	0.0000
## Question_19	0.0000	0.0000	0.0000	0.0000	0.0000	0.0146	0.0000
## Question_20	0.0000	0.0000	0.0172	0.0000	0.7655	0.0007	0.0000
## Question_21		0.0000	0.0118	0.0000	0.0000	0.0000	0.0000
## Question_22	0.0000		0.0000	0.6512	0.0000	0.0010	0.0000
## Question_23	0.0118	0.0000		0.0631	0.4869	0.0129	0.0000
## FAC1_1	0.0000	0.6512	0.0631		0.3654	0.7581	0.9397
## FAC2_1	0.0000	0.0000	0.4869	0.3654		0.5348	0.4812
## FAC3_1	0.0000	0.0010	0.0129	0.7581	0.5348		0.9497
## FAC4_1	0.0000	0.0000	0.0000	0.9397	0.4812	0.9497	
## FAC1_2	0.0000	0.3220	0.2750	0.0000	0.0087	0.0032	0.0314


```
## FAC2_2      0.0057      0.0000      0.0000      0.0131 0.2299 0.8805 0.0000
## FAC3_2      0.0000      0.0000      0.1520      0.0000 0.0000 0.5092 0.0036
## FAC4_2      0.0000      0.0004      0.0174      0.0002 0.0883 0.0000 0.8293
##            FAC1_2 FAC2_2 FAC3_2 FAC4_2
## id          0.0379 0.0010 0.1747 0.4811
## Question_01 0.0000 0.0002 0.0000 0.0000
## Question_02 0.0000 0.0000 0.0079 0.3236
## Question_03 0.0000 0.0000 0.0000 0.0000
## Question_04 0.0000 0.0011 0.0000 0.0000
## Question_05 0.0000 0.1163 0.0000 0.0000
## Question_06 0.0001 0.0032 0.0000 0.0000
## Question_07 0.0000 0.0006 0.0000 0.0000
## Question_08 0.0000 0.4671 0.0000 0.0000
## Question_09 0.0000 0.0000 0.0000 0.0145
## Question_10 0.0001 0.0002 0.0000 0.0000
## Question_11 0.0000 0.0000 0.0000 0.0000
## Question_12 0.0000 0.0015 0.0000 0.0000
## Question_13 0.0000 0.0043 0.0000 0.0000
## Question_14 0.0000 0.0002 0.0000 0.0000
## Question_15 0.0000 0.0000 0.0000 0.0000
## Question_16 0.0000 0.0000 0.0000 0.0000
## Question_17 0.0000 0.7104 0.0000 0.0000
## Question_18 0.0000 0.0461 0.0000 0.0000
## Question_19 0.0000 0.0000 0.0000 0.0003
## Question_20 0.0000 0.0000 0.0122 0.0000
## Question_21 0.0000 0.0057 0.0000 0.0000
## Question_22 0.3220 0.0000 0.0000 0.0004
## Question_23 0.2750 0.0000 0.1520 0.0174
## FAC1_1      0.0000 0.0131 0.0000 0.0002
## FAC2_1      0.0087 0.2299 0.0000 0.0883
## FAC3_1      0.0032 0.8805 0.5092 0.0000
## FAC4_1      0.0314 0.0000 0.0036 0.8293
## FAC1_2      0.9486 0.3488 0.7273
## FAC2_2      0.9486      0.4293 0.9823
## FAC3_2      0.3488 0.4293      0.4570
## FAC4_2      0.7273 0.9823 0.4570
```

```
library(corr)
```

```
df2_Flat_Cor_Mat_stretch <- df2_training %>%
  select(-id) %>% # remove ID variable since we don't need it
  correlate() %>% # calculate correlations
  stretch() %>% # make it tall
  fashion() # round it
```

```
## select: dropped one variable (id)
## Correlation computed with
## * Method: 'pearson'
## * Missing treated using: 'pairwise.complete.obs'
```

```
df2_Flat_Cor_Mat_stretch # This is running the flattened correlation matrix for easier readability
```

```
##           x           y           r
## 1 Question_01 Question_01
## 2 Question_01 Question_02 .09
## 3 Question_01 Question_03 .37
```

## 4	Question_01	Question_04	.48
## 5	Question_01	Question_05	.42
## 6	Question_01	Question_06	.26
## 7	Question_01	Question_07	.31
## 8	Question_01	Question_08	.34
## 9	Question_01	Question_09	-.08
## 10	Question_01	Question_10	.24
## 11	Question_01	Question_11	.34
## 12	Question_01	Question_12	.37
## 13	Question_01	Question_13	.37
## 14	Question_01	Question_14	.37
## 15	Question_01	Question_15	.25
## 16	Question_01	Question_16	.51
## 17	Question_01	Question_17	.39
## 18	Question_01	Question_18	.37
## 19	Question_01	Question_19	.20
## 20	Question_01	Question_20	.22
## 21	Question_01	Question_21	.36
## 22	Question_01	Question_22	-.11
## 23	Question_01	Question_23	-.03
## 24	Question_01	FAC1_1	.61
## 25	Question_01	FAC2_1	.23
## 26	Question_01	FAC3_1	.26
## 27	Question_01	FAC4_1	.05
## 28	Question_01	FAC1_2	.56
## 29	Question_01	FAC2_2	.10
## 30	Question_01	FAC3_2	.27
## 31	Question_01	FAC4_2	-.32
## 32	Question_02	Question_01	.09
## 33	Question_02	Question_02	
## 34	Question_02	Question_03	.33
## 35	Question_02	Question_04	.12
## 36	Question_02	Question_05	.13
## 37	Question_02	Question_06	.10
## 38	Question_02	Question_07	.17
## 39	Question_02	Question_08	.07
## 40	Question_02	Question_09	-.30
## 41	Question_02	Question_10	.09
## 42	Question_02	Question_11	.16
## 43	Question_02	Question_12	.19
## 44	Question_02	Question_13	.14
## 45	Question_02	Question_14	.20
## 46	Question_02	Question_15	.20
## 47	Question_02	Question_16	.19
## 48	Question_02	Question_17	.10
## 49	Question_02	Question_18	.18
## 50	Question_02	Question_19	.24
## 51	Question_02	Question_20	.18
## 52	Question_02	Question_21	.23
## 53	Question_02	Question_22	-.25
## 54	Question_02	Question_23	-.12
## 55	Question_02	FAC1_1	.21
## 56	Question_02	FAC2_1	.02
## 57	Question_02	FAC3_1	.01

## 58	Question_02	FAC4_1	-.59
## 59	Question_02	FAC1_2	.24
## 60	Question_02	FAC2_2	-.57
## 61	Question_02	FAC3_2	.07
## 62	Question_02	FAC4_2	-.03
## 63	Question_03	Question_01	.37
## 64	Question_03	Question_02	.33
## 65	Question_03	Question_03	
## 66	Question_03	Question_04	.37
## 67	Question_03	Question_05	.30
## 68	Question_03	Question_06	.26
## 69	Question_03	Question_07	.40
## 70	Question_03	Question_08	.27
## 71	Question_03	Question_09	-.34
## 72	Question_03	Question_10	.21
## 73	Question_03	Question_11	.36
## 74	Question_03	Question_12	.40
## 75	Question_03	Question_13	.32
## 76	Question_03	Question_14	.39
## 77	Question_03	Question_15	.33
## 78	Question_03	Question_16	.41
## 79	Question_03	Question_17	.33
## 80	Question_03	Question_18	.36
## 81	Question_03	Question_19	.36
## 82	Question_03	Question_20	.33
## 83	Question_03	Question_21	.41
## 84	Question_03	Question_22	-.24
## 85	Question_03	Question_23	-.18
## 86	Question_03	FAC1_1	.54
## 87	Question_03	FAC2_1	.17
## 88	Question_03	FAC3_1	.16
## 89	Question_03	FAC4_1	-.50
## 90	Question_03	FAC1_2	.53
## 91	Question_03	FAC2_2	-.45
## 92	Question_03	FAC3_2	.24
## 93	Question_03	FAC4_2	-.22
## 94	Question_04	Question_01	.48
## 95	Question_04	Question_02	.12
## 96	Question_04	Question_03	.37
## 97	Question_04	Question_04	
## 98	Question_04	Question_05	.41
## 99	Question_04	Question_06	.29
## 100	Question_04	Question_07	.41
## 101	Question_04	Question_08	.37
## 102	Question_04	Question_09	-.09
## 103	Question_04	Question_10	.23
## 104	Question_04	Question_11	.38
## 105	Question_04	Question_12	.44
## 106	Question_04	Question_13	.34
## 107	Question_04	Question_14	.37
## 108	Question_04	Question_15	.33
## 109	Question_04	Question_16	.41
## 110	Question_04	Question_17	.39
## 111	Question_04	Question_18	.39

```

## 112 Question_04 Question_19 .19
## 113 Question_04 Question_20 .25
## 114 Question_04 Question_21 .43
## 115 Question_04 Question_22 -.12
## 116 Question_04 Question_23 -.02
## 117 Question_04 FAC1_1 .62
## 118 Question_04 FAC2_1 .28
## 119 Question_04 FAC3_1 .26
## 120 Question_04 FAC4_1 .04
## 121 Question_04 FAC1_2 .56
## 122 Question_04 FAC2_2 .09
## 123 Question_04 FAC3_2 .32
## 124 Question_04 FAC4_2 -.32
## 125 Question_05 Question_01 .42
## 126 Question_05 Question_02 .13
## 127 Question_05 Question_03 .30
## 128 Question_05 Question_04 .41
## 129 Question_05 Question_05
## 130 Question_05 Question_06 .26
## 131 Question_05 Question_07 .32
## 132 Question_05 Question_08 .27
## 133 Question_05 Question_09 -.09
## 134 Question_05 Question_10 .29
## 135 Question_05 Question_11 .28
## 136 Question_05 Question_12 .34
## 137 Question_05 Question_13 .31
## 138 Question_05 Question_14 .32
## 139 Question_05 Question_15 .26
## 140 Question_05 Question_16 .39
## 141 Question_05 Question_17 .32
## 142 Question_05 Question_18 .31
## 143 Question_05 Question_19 .16
## 144 Question_05 Question_20 .20
## 145 Question_05 Question_21 .33
## 146 Question_05 Question_22 -.14
## 147 Question_05 Question_23 -.04
## 148 Question_05 FAC1_1 .50
## 149 Question_05 FAC2_1 .26
## 150 Question_05 FAC3_1 .18
## 151 Question_05 FAC4_1 -.00
## 152 Question_05 FAC1_2 .46
## 153 Question_05 FAC2_2 .04
## 154 Question_05 FAC3_2 .30
## 155 Question_05 FAC4_2 -.23
## 156 Question_06 Question_01 .26
## 157 Question_06 Question_02 .10
## 158 Question_06 Question_03 .26
## 159 Question_06 Question_04 .29
## 160 Question_06 Question_05 .26
## 161 Question_06 Question_06
## 162 Question_06 Question_07 .51
## 163 Question_06 Question_08 .24
## 164 Question_06 Question_09 -.14
## 165 Question_06 Question_10 .31

```

## 166	Question_06	Question_11	.35
## 167	Question_06	Question_12	.34
## 168	Question_06	Question_13	.47
## 169	Question_06	Question_14	.40
## 170	Question_06	Question_15	.34
## 171	Question_06	Question_16	.27
## 172	Question_06	Question_17	.31
## 173	Question_06	Question_18	.50
## 174	Question_06	Question_19	.22
## 175	Question_06	Question_20	.11
## 176	Question_06	Question_21	.27
## 177	Question_06	Question_22	-.17
## 178	Question_06	Question_23	-.07
## 179	Question_06	FAC1_1	-.02
## 180	Question_06	FAC2_1	.88
## 181	Question_06	FAC3_1	.14
## 182	Question_06	FAC4_1	-.13
## 183	Question_06	FAC1_2	-.11
## 184	Question_06	FAC2_2	-.08
## 185	Question_06	FAC3_2	.87
## 186	Question_06	FAC4_2	-.16
## 187	Question_07	Question_01	.31
## 188	Question_07	Question_02	.17
## 189	Question_07	Question_03	.40
## 190	Question_07	Question_04	.41
## 191	Question_07	Question_05	.32
## 192	Question_07	Question_06	.51
## 193	Question_07	Question_07	
## 194	Question_07	Question_08	.27
## 195	Question_07	Question_09	-.15
## 196	Question_07	Question_10	.29
## 197	Question_07	Question_11	.34
## 198	Question_07	Question_12	.44
## 199	Question_07	Question_13	.43
## 200	Question_07	Question_14	.45
## 201	Question_07	Question_15	.41
## 202	Question_07	Question_16	.38
## 203	Question_07	Question_17	.38
## 204	Question_07	Question_18	.50
## 205	Question_07	Question_19	.31
## 206	Question_07	Question_20	.25
## 207	Question_07	Question_21	.51
## 208	Question_07	Question_22	-.20
## 209	Question_07	Question_23	-.08
## 210	Question_07	FAC1_1	.39
## 211	Question_07	FAC2_1	.63
## 212	Question_07	FAC3_1	.12
## 213	Question_07	FAC4_1	-.15
## 214	Question_07	FAC1_2	.32
## 215	Question_07	FAC2_2	-.10
## 216	Question_07	FAC3_2	.66
## 217	Question_07	FAC4_2	-.17
## 218	Question_08	Question_01	.34
## 219	Question_08	Question_02	.07

##	220	Question_08	Question_03	.27
##	221	Question_08	Question_04	.37
##	222	Question_08	Question_05	.27
##	223	Question_08	Question_06	.24
##	224	Question_08	Question_07	.27
##	225	Question_08	Question_08	
##	226	Question_08	Question_09	.02
##	227	Question_08	Question_10	.18
##	228	Question_08	Question_11	.63
##	229	Question_08	Question_12	.28
##	230	Question_08	Question_13	.33
##	231	Question_08	Question_14	.31
##	232	Question_08	Question_15	.30
##	233	Question_08	Question_16	.32
##	234	Question_08	Question_17	.58
##	235	Question_08	Question_18	.32
##	236	Question_08	Question_19	.17
##	237	Question_08	Question_20	.20
##	238	Question_08	Question_21	.29
##	239	Question_08	Question_22	-.09
##	240	Question_08	Question_23	-.04
##	241	Question_08	FAC1_1	.21
##	242	Question_08	FAC2_1	.15
##	243	Question_08	FAC3_1	.86
##	244	Question_08	FAC4_1	.00
##	245	Question_08	FAC1_2	.12
##	246	Question_08	FAC2_2	.02
##	247	Question_08	FAC3_2	.14
##	248	Question_08	FAC4_2	-.88
##	249	Question_09	Question_01	-.08
##	250	Question_09	Question_02	-.30
##	251	Question_09	Question_03	-.34
##	252	Question_09	Question_04	-.09
##	253	Question_09	Question_05	-.09
##	254	Question_09	Question_06	-.14
##	255	Question_09	Question_07	-.15
##	256	Question_09	Question_08	.02
##	257	Question_09	Question_09	
##	258	Question_09	Question_10	-.15
##	259	Question_09	Question_11	-.11
##	260	Question_09	Question_12	-.18
##	261	Question_09	Question_13	-.16
##	262	Question_09	Question_14	-.14
##	263	Question_09	Question_15	-.20
##	264	Question_09	Question_16	-.19
##	265	Question_09	Question_17	-.03
##	266	Question_09	Question_18	-.16
##	267	Question_09	Question_19	-.27
##	268	Question_09	Question_20	-.15
##	269	Question_09	Question_21	-.15
##	270	Question_09	Question_22	.28
##	271	Question_09	Question_23	.21
##	272	Question_09	FAC1_1	-.10
##	273	Question_09	FAC2_1	-.08

```

## 274 Question_09      FAC3_1  .08
## 275 Question_09      FAC4_1  .73
## 276 Question_09      FAC1_2 -.14
## 277 Question_09      FAC2_2  .72
## 278 Question_09      FAC3_2 -.14
## 279 Question_09      FAC4_2 -.07
## 280 Question_10 Question_01  .24
## 281 Question_10 Question_02  .09
## 282 Question_10 Question_03  .21
## 283 Question_10 Question_04  .23
## 284 Question_10 Question_05  .29
## 285 Question_10 Question_06  .31
## 286 Question_10 Question_07  .29
## 287 Question_10 Question_08  .18
## 288 Question_10 Question_09 -.15
## 289 Question_10 Question_10
## 290 Question_10 Question_11  .28
## 291 Question_10 Question_12  .26
## 292 Question_10 Question_13  .29
## 293 Question_10 Question_14  .24
## 294 Question_10 Question_15  .32
## 295 Question_10 Question_16  .30
## 296 Question_10 Question_17  .22
## 297 Question_10 Question_18  .30
## 298 Question_10 Question_19  .16
## 299 Question_10 Question_20  .12
## 300 Question_10 Question_21  .20
## 301 Question_10 Question_22 -.12
## 302 Question_10 Question_23 -.04
## 303 Question_10      FAC1_1  .15
## 304 Question_10      FAC2_1  .41
## 305 Question_10      FAC3_1  .14
## 306 Question_10      FAC4_1 -.13
## 307 Question_10      FAC1_2  .11
## 308 Question_10      FAC2_2 -.10
## 309 Question_10      FAC3_2  .43
## 310 Question_10      FAC4_2 -.17
## 311 Question_11 Question_01  .34
## 312 Question_11 Question_02  .16
## 313 Question_11 Question_03  .36
## 314 Question_11 Question_04  .38
## 315 Question_11 Question_05  .28
## 316 Question_11 Question_06  .35
## 317 Question_11 Question_07  .34
## 318 Question_11 Question_08  .63
## 319 Question_11 Question_09 -.11
## 320 Question_11 Question_10  .28
## 321 Question_11 Question_11
## 322 Question_11 Question_12  .37
## 323 Question_11 Question_13  .40
## 324 Question_11 Question_14  .33
## 325 Question_11 Question_15  .36
## 326 Question_11 Question_16  .36
## 327 Question_11 Question_17  .57

```

```

## 328 Question_11 Question_18 .38
## 329 Question_11 Question_19 .25
## 330 Question_11 Question_20 .28
## 331 Question_11 Question_21 .36
## 332 Question_11 Question_22 -.19
## 333 Question_11 Question_23 -.08
## 334 Question_11 FAC1_1 .20
## 335 Question_11 FAC2_1 .27
## 336 Question_11 FAC3_1 .78
## 337 Question_11 FAC4_1 -.22
## 338 Question_11 FAC1_2 .12
## 339 Question_11 FAC2_2 -.20
## 340 Question_11 FAC3_2 .27
## 341 Question_11 FAC4_2 -.80
## 342 Question_12 Question_01 .37
## 343 Question_12 Question_02 .19
## 344 Question_12 Question_03 .40
## 345 Question_12 Question_04 .44
## 346 Question_12 Question_05 .34
## 347 Question_12 Question_06 .34
## 348 Question_12 Question_07 .44
## 349 Question_12 Question_08 .28
## 350 Question_12 Question_09 -.18
## 351 Question_12 Question_10 .26
## 352 Question_12 Question_11 .37
## 353 Question_12 Question_12
## 354 Question_12 Question_13 .51
## 355 Question_12 Question_14 .45
## 356 Question_12 Question_15 .35
## 357 Question_12 Question_16 .41
## 358 Question_12 Question_17 .33
## 359 Question_12 Question_18 .51
## 360 Question_12 Question_19 .31
## 361 Question_12 Question_20 .29
## 362 Question_12 Question_21 .45
## 363 Question_12 Question_22 -.19
## 364 Question_12 Question_23 -.05
## 365 Question_12 FAC1_1 .56
## 366 Question_12 FAC2_1 .44
## 367 Question_12 FAC3_1 .10
## 368 Question_12 FAC4_1 -.15
## 369 Question_12 FAC1_2 .51
## 370 Question_12 FAC2_2 -.09
## 371 Question_12 FAC3_2 .49
## 372 Question_12 FAC4_2 -.17
## 373 Question_13 Question_01 .37
## 374 Question_13 Question_02 .14
## 375 Question_13 Question_03 .32
## 376 Question_13 Question_04 .34
## 377 Question_13 Question_05 .31
## 378 Question_13 Question_06 .47
## 379 Question_13 Question_07 .43
## 380 Question_13 Question_08 .33
## 381 Question_13 Question_09 -.16

```



```

## 382 Question_13 Question_10 .29
## 383 Question_13 Question_11 .40
## 384 Question_13 Question_12 .51
## 385 Question_13 Question_13
## 386 Question_13 Question_14 .45
## 387 Question_13 Question_15 .35
## 388 Question_13 Question_16 .35
## 389 Question_13 Question_17 .40
## 390 Question_13 Question_18 .52
## 391 Question_13 Question_19 .27
## 392 Question_13 Question_20 .21
## 393 Question_13 Question_21 .36
## 394 Question_13 Question_22 -.17
## 395 Question_13 Question_23 -.03
## 396 Question_13 FAC1_1 .28
## 397 Question_13 FAC2_1 .63
## 398 Question_13 FAC3_1 .22
## 399 Question_13 FAC4_1 -.13
## 400 Question_13 FAC1_2 .21
## 401 Question_13 FAC2_2 -.08
## 402 Question_13 FAC3_2 .65
## 403 Question_13 FAC4_2 -.27
## 404 Question_14 Question_01 .37
## 405 Question_14 Question_02 .20
## 406 Question_14 Question_03 .39
## 407 Question_14 Question_04 .37
## 408 Question_14 Question_05 .32
## 409 Question_14 Question_06 .40
## 410 Question_14 Question_07 .45
## 411 Question_14 Question_08 .31
## 412 Question_14 Question_09 -.14
## 413 Question_14 Question_10 .24
## 414 Question_14 Question_11 .33
## 415 Question_14 Question_12 .45
## 416 Question_14 Question_13 .45
## 417 Question_14 Question_14
## 418 Question_14 Question_15 .37
## 419 Question_14 Question_16 .43
## 420 Question_14 Question_17 .36
## 421 Question_14 Question_18 .51
## 422 Question_14 Question_19 .30
## 423 Question_14 Question_20 .26
## 424 Question_14 Question_21 .40
## 425 Question_14 Question_22 -.23
## 426 Question_14 Question_23 -.09
## 427 Question_14 FAC1_1 .43
## 428 Question_14 FAC2_1 .52
## 429 Question_14 FAC3_1 .14
## 430 Question_14 FAC4_1 -.16
## 431 Question_14 FAC1_2 .38
## 432 Question_14 FAC2_2 -.10
## 433 Question_14 FAC3_2 .56
## 434 Question_14 FAC4_2 -.20
## 435 Question_15 Question_01 .25

```

```

## 436 Question_15 Question_02 .20
## 437 Question_15 Question_03 .33
## 438 Question_15 Question_04 .33
## 439 Question_15 Question_05 .26
## 440 Question_15 Question_06 .34
## 441 Question_15 Question_07 .41
## 442 Question_15 Question_08 .30
## 443 Question_15 Question_09 -.20
## 444 Question_15 Question_10 .32
## 445 Question_15 Question_11 .36
## 446 Question_15 Question_12 .35
## 447 Question_15 Question_13 .35
## 448 Question_15 Question_14 .37
## 449 Question_15 Question_15
## 450 Question_15 Question_16 .46
## 451 Question_15 Question_17 .37
## 452 Question_15 Question_18 .35
## 453 Question_15 Question_19 .25
## 454 Question_15 Question_20 .23
## 455 Question_15 Question_21 .30
## 456 Question_15 Question_22 -.19
## 457 Question_15 Question_23 -.06
## 458 Question_15 FAC1_1 .29
## 459 Question_15 FAC2_1 .39
## 460 Question_15 FAC3_1 .25
## 461 Question_15 FAC4_1 -.26
## 462 Question_15 FAC1_2 .24
## 463 Question_15 FAC2_2 -.22
## 464 Question_15 FAC3_2 .42
## 465 Question_15 FAC4_2 -.29
## 466 Question_16 Question_01 .51
## 467 Question_16 Question_02 .19
## 468 Question_16 Question_03 .41
## 469 Question_16 Question_04 .41
## 470 Question_16 Question_05 .39
## 471 Question_16 Question_06 .27
## 472 Question_16 Question_07 .38
## 473 Question_16 Question_08 .32
## 474 Question_16 Question_09 -.19
## 475 Question_16 Question_10 .30
## 476 Question_16 Question_11 .36
## 477 Question_16 Question_12 .41
## 478 Question_16 Question_13 .35
## 479 Question_16 Question_14 .43
## 480 Question_16 Question_15 .46
## 481 Question_16 Question_16
## 482 Question_16 Question_17 .40
## 483 Question_16 Question_18 .41
## 484 Question_16 Question_19 .27
## 485 Question_16 Question_20 .29
## 486 Question_16 Question_21 .42
## 487 Question_16 Question_22 -.20
## 488 Question_16 Question_23 -.11
## 489 Question_16 FAC1_1 .61

```

```

## 490 Question_16      FAC2_1  .27
## 491 Question_16      FAC3_1  .22
## 492 Question_16      FAC4_1  -.16
## 493 Question_16      FAC1_2  .57
## 494 Question_16      FAC2_2  -.11
## 495 Question_16      FAC3_2  .32
## 496 Question_16      FAC4_2  -.29
## 497 Question_17 Question_01  .39
## 498 Question_17 Question_02  .10
## 499 Question_17 Question_03  .33
## 500 Question_17 Question_04  .39
## 501 Question_17 Question_05  .32
## 502 Question_17 Question_06  .31
## 503 Question_17 Question_07  .38
## 504 Question_17 Question_08  .58
## 505 Question_17 Question_09  -.03
## 506 Question_17 Question_10  .22
## 507 Question_17 Question_11  .57
## 508 Question_17 Question_12  .33
## 509 Question_17 Question_13  .40
## 510 Question_17 Question_14  .36
## 511 Question_17 Question_15  .37
## 512 Question_17 Question_16  .40
## 513 Question_17 Question_17
## 514 Question_17 Question_18  .39
## 515 Question_17 Question_19  .17
## 516 Question_17 Question_20  .21
## 517 Question_17 Question_21  .34
## 518 Question_17 Question_22  -.15
## 519 Question_17 Question_23  -.08
## 520 Question_17      FAC1_1  .29
## 521 Question_17      FAC2_1  .28
## 522 Question_17      FAC3_1  .71
## 523 Question_17      FAC4_1  -.04
## 524 Question_17      FAC1_2  .20
## 525 Question_17      FAC2_2  -.01
## 526 Question_17      FAC3_2  .29
## 527 Question_17      FAC4_2  -.74
## 528 Question_18 Question_01  .37
## 529 Question_18 Question_02  .18
## 530 Question_18 Question_03  .36
## 531 Question_18 Question_04  .39
## 532 Question_18 Question_05  .31
## 533 Question_18 Question_06  .50
## 534 Question_18 Question_07  .50
## 535 Question_18 Question_08  .32
## 536 Question_18 Question_09  -.16
## 537 Question_18 Question_10  .30
## 538 Question_18 Question_11  .38
## 539 Question_18 Question_12  .51
## 540 Question_18 Question_13  .52
## 541 Question_18 Question_14  .51
## 542 Question_18 Question_15  .35
## 543 Question_18 Question_16  .41

```

```

## 544 Question_18 Question_17 .39
## 545 Question_18 Question_18
## 546 Question_18 Question_19 .28
## 547 Question_18 Question_20 .26
## 548 Question_18 Question_21 .43
## 549 Question_18 Question_22 -.16
## 550 Question_18 Question_23 -.07
## 551 Question_18 FAC1_1 .38
## 552 Question_18 FAC2_1 .68
## 553 Question_18 FAC3_1 .14
## 554 Question_18 FAC4_1 -.12
## 555 Question_18 FAC1_2 .31
## 556 Question_18 FAC2_2 -.06
## 557 Question_18 FAC3_2 .71
## 558 Question_18 FAC4_2 -.19
## 559 Question_19 Question_01 .20
## 560 Question_19 Question_02 .24
## 561 Question_19 Question_03 .36
## 562 Question_19 Question_04 .19
## 563 Question_19 Question_05 .16
## 564 Question_19 Question_06 .22
## 565 Question_19 Question_07 .31
## 566 Question_19 Question_08 .17
## 567 Question_19 Question_09 -.27
## 568 Question_19 Question_10 .16
## 569 Question_19 Question_11 .25
## 570 Question_19 Question_12 .31
## 571 Question_19 Question_13 .27
## 572 Question_19 Question_14 .30
## 573 Question_19 Question_15 .25
## 574 Question_19 Question_16 .27
## 575 Question_19 Question_17 .17
## 576 Question_19 Question_18 .28
## 577 Question_19 Question_19
## 578 Question_19 Question_20 .26
## 579 Question_19 Question_21 .28
## 580 Question_19 Question_22 -.27
## 581 Question_19 Question_23 -.14
## 582 Question_19 FAC1_1 .28
## 583 Question_19 FAC2_1 .19
## 584 Question_19 FAC3_1 .07
## 585 Question_19 FAC4_1 -.49
## 586 Question_19 FAC1_2 .28
## 587 Question_19 FAC2_2 -.46
## 588 Question_19 FAC3_2 .24
## 589 Question_19 FAC4_2 -.10
## 590 Question_20 Question_01 .22
## 591 Question_20 Question_02 .18
## 592 Question_20 Question_03 .33
## 593 Question_20 Question_04 .25
## 594 Question_20 Question_05 .20
## 595 Question_20 Question_06 .11
## 596 Question_20 Question_07 .25
## 597 Question_20 Question_08 .20

```

```

## 598 Question_20 Question_09 -.15
## 599 Question_20 Question_10 .12
## 600 Question_20 Question_11 .28
## 601 Question_20 Question_12 .29
## 602 Question_20 Question_13 .21
## 603 Question_20 Question_14 .26
## 604 Question_20 Question_15 .23
## 605 Question_20 Question_16 .29
## 606 Question_20 Question_17 .21
## 607 Question_20 Question_18 .26
## 608 Question_20 Question_19 .26
## 609 Question_20 Question_20
## 610 Question_20 Question_21 .50
## 611 Question_20 Question_22 -.12
## 612 Question_20 Question_23 -.07
## 613 Question_20 FAC1_1 .55
## 614 Question_20 FAC2_1 .01
## 615 Question_20 FAC3_1 .09
## 616 Question_20 FAC4_1 -.22
## 617 Question_20 FAC1_2 .55
## 618 Question_20 FAC2_2 -.19
## 619 Question_20 FAC3_2 .07
## 620 Question_20 FAC4_2 -.15
## 621 Question_21 Question_01 .36
## 622 Question_21 Question_02 .23
## 623 Question_21 Question_03 .41
## 624 Question_21 Question_04 .43
## 625 Question_21 Question_05 .33
## 626 Question_21 Question_06 .27
## 627 Question_21 Question_07 .51
## 628 Question_21 Question_08 .29
## 629 Question_21 Question_09 -.15
## 630 Question_21 Question_10 .20
## 631 Question_21 Question_11 .36
## 632 Question_21 Question_12 .45
## 633 Question_21 Question_13 .36
## 634 Question_21 Question_14 .40
## 635 Question_21 Question_15 .30
## 636 Question_21 Question_16 .42
## 637 Question_21 Question_17 .34
## 638 Question_21 Question_18 .43
## 639 Question_21 Question_19 .28
## 640 Question_21 Question_20 .50
## 641 Question_21 Question_21
## 642 Question_21 Question_22 -.13
## 643 Question_21 Question_23 -.07
## 644 Question_21 FAC1_1 .70
## 645 Question_21 FAC2_1 .25
## 646 Question_21 FAC3_1 .12
## 647 Question_21 FAC4_1 -.14
## 648 Question_21 FAC1_2 .66
## 649 Question_21 FAC2_2 -.08
## 650 Question_21 FAC3_2 .32
## 651 Question_21 FAC4_2 -.19

```

```

## 652 Question_22 Question_01 -.11
## 653 Question_22 Question_02 -.25
## 654 Question_22 Question_03 -.24
## 655 Question_22 Question_04 -.12
## 656 Question_22 Question_05 -.14
## 657 Question_22 Question_06 -.17
## 658 Question_22 Question_07 -.20
## 659 Question_22 Question_08 -.09
## 660 Question_22 Question_09 .28
## 661 Question_22 Question_10 -.12
## 662 Question_22 Question_11 -.19
## 663 Question_22 Question_12 -.19
## 664 Question_22 Question_13 -.17
## 665 Question_22 Question_14 -.23
## 666 Question_22 Question_15 -.19
## 667 Question_22 Question_16 -.20
## 668 Question_22 Question_17 -.15
## 669 Question_22 Question_18 -.16
## 670 Question_22 Question_19 -.27
## 671 Question_22 Question_20 -.12
## 672 Question_22 Question_21 -.13
## 673 Question_22 Question_22
## 674 Question_22 Question_23 .25
## 675 Question_22 FAC1_1 -.01
## 676 Question_22 FAC2_1 -.16
## 677 Question_22 FAC3_1 -.09
## 678 Question_22 FAC4_1 .60
## 679 Question_22 FAC1_2 -.03
## 680 Question_22 FAC2_2 .59
## 681 Question_22 FAC3_2 -.19
## 682 Question_22 FAC4_2 .10
## 683 Question_23 Question_01 -.03
## 684 Question_23 Question_02 -.12
## 685 Question_23 Question_03 -.18
## 686 Question_23 Question_04 -.02
## 687 Question_23 Question_05 -.04
## 688 Question_23 Question_06 -.07
## 689 Question_23 Question_07 -.08
## 690 Question_23 Question_08 -.04
## 691 Question_23 Question_09 .21
## 692 Question_23 Question_10 -.04
## 693 Question_23 Question_11 -.08
## 694 Question_23 Question_12 -.05
## 695 Question_23 Question_13 -.03
## 696 Question_23 Question_14 -.09
## 697 Question_23 Question_15 -.06
## 698 Question_23 Question_16 -.11
## 699 Question_23 Question_17 -.08
## 700 Question_23 Question_18 -.07
## 701 Question_23 Question_19 -.14
## 702 Question_23 Question_20 -.07
## 703 Question_23 Question_21 -.07
## 704 Question_23 Question_22 .25
## 705 Question_23 Question_23

```

## 706	Question_23	FAC1_1	.05
## 707	Question_23	FAC2_1	-.02
## 708	Question_23	FAC3_1	-.07
## 709	Question_23	FAC4_1	.45
## 710	Question_23	FAC1_2	.03
## 711	Question_23	FAC2_2	.45
## 712	Question_23	FAC3_2	-.04
## 713	Question_23	FAC4_2	.07
## 714	FAC1_1	Question_01	.61
## 715	FAC1_1	Question_02	.21
## 716	FAC1_1	Question_03	.54
## 717	FAC1_1	Question_04	.62
## 718	FAC1_1	Question_05	.50
## 719	FAC1_1	Question_06	-.02
## 720	FAC1_1	Question_07	.39
## 721	FAC1_1	Question_08	.21
## 722	FAC1_1	Question_09	-.10
## 723	FAC1_1	Question_10	.15
## 724	FAC1_1	Question_11	.20
## 725	FAC1_1	Question_12	.56
## 726	FAC1_1	Question_13	.28
## 727	FAC1_1	Question_14	.43
## 728	FAC1_1	Question_15	.29
## 729	FAC1_1	Question_16	.61
## 730	FAC1_1	Question_17	.29
## 731	FAC1_1	Question_18	.38
## 732	FAC1_1	Question_19	.28
## 733	FAC1_1	Question_20	.55
## 734	FAC1_1	Question_21	.70
## 735	FAC1_1	Question_22	-.01
## 736	FAC1_1	Question_23	.05
## 737	FAC1_1	FAC1_1	
## 738	FAC1_1	FAC2_1	.03
## 739	FAC1_1	FAC3_1	.01
## 740	FAC1_1	FAC4_1	.00
## 741	FAC1_1	FAC1_2	.99
## 742	FAC1_1	FAC2_2	.07
## 743	FAC1_1	FAC3_2	.12
## 744	FAC1_1	FAC4_2	-.10
## 745	FAC2_1	Question_01	.23
## 746	FAC2_1	Question_02	.02
## 747	FAC2_1	Question_03	.17
## 748	FAC2_1	Question_04	.28
## 749	FAC2_1	Question_05	.26
## 750	FAC2_1	Question_06	.88
## 751	FAC2_1	Question_07	.63
## 752	FAC2_1	Question_08	.15
## 753	FAC2_1	Question_09	-.08
## 754	FAC2_1	Question_10	.41
## 755	FAC2_1	Question_11	.27
## 756	FAC2_1	Question_12	.44
## 757	FAC2_1	Question_13	.63
## 758	FAC2_1	Question_14	.52
## 759	FAC2_1	Question_15	.39

## 760	FAC2_1	Question_16	.27
## 761	FAC2_1	Question_17	.28
## 762	FAC2_1	Question_18	.68
## 763	FAC2_1	Question_19	.19
## 764	FAC2_1	Question_20	.01
## 765	FAC2_1	Question_21	.25
## 766	FAC2_1	Question_22	-.16
## 767	FAC2_1	Question_23	-.02
## 768	FAC2_1	FAC1_1	.03
## 769	FAC2_1	FAC2_1	
## 770	FAC2_1	FAC3_1	.02
## 771	FAC2_1	FAC4_1	-.02
## 772	FAC2_1	FAC1_2	-.07
## 773	FAC2_1	FAC2_2	.03
## 774	FAC2_1	FAC3_2	.99
## 775	FAC2_1	FAC4_2	-.05
## 776	FAC3_1	Question_01	.26
## 777	FAC3_1	Question_02	.01
## 778	FAC3_1	Question_03	.16
## 779	FAC3_1	Question_04	.26
## 780	FAC3_1	Question_05	.18
## 781	FAC3_1	Question_06	.14
## 782	FAC3_1	Question_07	.12
## 783	FAC3_1	Question_08	.86
## 784	FAC3_1	Question_09	.08
## 785	FAC3_1	Question_10	.14
## 786	FAC3_1	Question_11	.78
## 787	FAC3_1	Question_12	.10
## 788	FAC3_1	Question_13	.22
## 789	FAC3_1	Question_14	.14
## 790	FAC3_1	Question_15	.25
## 791	FAC3_1	Question_16	.22
## 792	FAC3_1	Question_17	.71
## 793	FAC3_1	Question_18	.14
## 794	FAC3_1	Question_19	.07
## 795	FAC3_1	Question_20	.09
## 796	FAC3_1	Question_21	.12
## 797	FAC3_1	Question_22	-.09
## 798	FAC3_1	Question_23	-.07
## 799	FAC3_1	FAC1_1	.01
## 800	FAC3_1	FAC2_1	.02
## 801	FAC3_1	FAC3_1	
## 802	FAC3_1	FAC4_1	-.00
## 803	FAC3_1	FAC1_2	-.08
## 804	FAC3_1	FAC2_2	-.00
## 805	FAC3_1	FAC3_2	-.02
## 806	FAC3_1	FAC4_2	-1.00
## 807	FAC4_1	Question_01	.05
## 808	FAC4_1	Question_02	-.59
## 809	FAC4_1	Question_03	-.50
## 810	FAC4_1	Question_04	.04
## 811	FAC4_1	Question_05	-.00
## 812	FAC4_1	Question_06	-.13
## 813	FAC4_1	Question_07	-.15

## 814	FAC4_1	Question_08	.00
## 815	FAC4_1	Question_09	.73
## 816	FAC4_1	Question_10	-.13
## 817	FAC4_1	Question_11	-.22
## 818	FAC4_1	Question_12	-.15
## 819	FAC4_1	Question_13	-.13
## 820	FAC4_1	Question_14	-.16
## 821	FAC4_1	Question_15	-.26
## 822	FAC4_1	Question_16	-.16
## 823	FAC4_1	Question_17	-.04
## 824	FAC4_1	Question_18	-.12
## 825	FAC4_1	Question_19	-.49
## 826	FAC4_1	Question_20	-.22
## 827	FAC4_1	Question_21	-.14
## 828	FAC4_1	Question_22	.60
## 829	FAC4_1	Question_23	.45
## 830	FAC4_1	FAC1_1	.00
## 831	FAC4_1	FAC2_1	-.02
## 832	FAC4_1	FAC3_1	-.00
## 833	FAC4_1	FAC4_1	
## 834	FAC4_1	FAC1_2	-.06
## 835	FAC4_1	FAC2_2	1.00
## 836	FAC4_1	FAC3_2	-.08
## 837	FAC4_1	FAC4_2	.01
## 838	FAC1_2	Question_01	.56
## 839	FAC1_2	Question_02	.24
## 840	FAC1_2	Question_03	.53
## 841	FAC1_2	Question_04	.56
## 842	FAC1_2	Question_05	.46
## 843	FAC1_2	Question_06	-.11
## 844	FAC1_2	Question_07	.32
## 845	FAC1_2	Question_08	.12
## 846	FAC1_2	Question_09	-.14
## 847	FAC1_2	Question_10	.11
## 848	FAC1_2	Question_11	.12
## 849	FAC1_2	Question_12	.51
## 850	FAC1_2	Question_13	.21
## 851	FAC1_2	Question_14	.38
## 852	FAC1_2	Question_15	.24
## 853	FAC1_2	Question_16	.57
## 854	FAC1_2	Question_17	.20
## 855	FAC1_2	Question_18	.31
## 856	FAC1_2	Question_19	.28
## 857	FAC1_2	Question_20	.55
## 858	FAC1_2	Question_21	.66
## 859	FAC1_2	Question_22	-.03
## 860	FAC1_2	Question_23	.03
## 861	FAC1_2	FAC1_1	.99
## 862	FAC1_2	FAC2_1	-.07
## 863	FAC1_2	FAC3_1	-.08
## 864	FAC1_2	FAC4_1	-.06
## 865	FAC1_2	FAC1_2	
## 866	FAC1_2	FAC2_2	.00
## 867	FAC1_2	FAC3_2	.03

## 868	FAC1_2	FAC4_2	-.01
## 869	FAC2_2	Question_01	.10
## 870	FAC2_2	Question_02	-.57
## 871	FAC2_2	Question_03	-.45
## 872	FAC2_2	Question_04	.09
## 873	FAC2_2	Question_05	.04
## 874	FAC2_2	Question_06	-.08
## 875	FAC2_2	Question_07	-.10
## 876	FAC2_2	Question_08	.02
## 877	FAC2_2	Question_09	.72
## 878	FAC2_2	Question_10	-.10
## 879	FAC2_2	Question_11	-.20
## 880	FAC2_2	Question_12	-.09
## 881	FAC2_2	Question_13	-.08
## 882	FAC2_2	Question_14	-.10
## 883	FAC2_2	Question_15	-.22
## 884	FAC2_2	Question_16	-.11
## 885	FAC2_2	Question_17	-.01
## 886	FAC2_2	Question_18	-.06
## 887	FAC2_2	Question_19	-.46
## 888	FAC2_2	Question_20	-.19
## 889	FAC2_2	Question_21	-.08
## 890	FAC2_2	Question_22	.59
## 891	FAC2_2	Question_23	.45
## 892	FAC2_2	FAC1_1	.07
## 893	FAC2_2	FAC2_1	.03
## 894	FAC2_2	FAC3_1	-.00
## 895	FAC2_2	FAC4_1	1.00
## 896	FAC2_2	FAC1_2	.00
## 897	FAC2_2	FAC2_2	
## 898	FAC2_2	FAC3_2	-.02
## 899	FAC2_2	FAC4_2	.00
## 900	FAC3_2	Question_01	.27
## 901	FAC3_2	Question_02	.07
## 902	FAC3_2	Question_03	.24
## 903	FAC3_2	Question_04	.32
## 904	FAC3_2	Question_05	.30
## 905	FAC3_2	Question_06	.87
## 906	FAC3_2	Question_07	.66
## 907	FAC3_2	Question_08	.14
## 908	FAC3_2	Question_09	-.14
## 909	FAC3_2	Question_10	.43
## 910	FAC3_2	Question_11	.27
## 911	FAC3_2	Question_12	.49
## 912	FAC3_2	Question_13	.65
## 913	FAC3_2	Question_14	.56
## 914	FAC3_2	Question_15	.42
## 915	FAC3_2	Question_16	.32
## 916	FAC3_2	Question_17	.29
## 917	FAC3_2	Question_18	.71
## 918	FAC3_2	Question_19	.24
## 919	FAC3_2	Question_20	.07
## 920	FAC3_2	Question_21	.32
## 921	FAC3_2	Question_22	-.19

```
## 922     FAC3_2 Question_23  -.04
## 923     FAC3_2     FAC1_1   .12
## 924     FAC3_2     FAC2_1   .99
## 925     FAC3_2     FAC3_1  -.02
## 926     FAC3_2     FAC4_1  -.08
## 927     FAC3_2     FAC1_2   .03
## 928     FAC3_2     FAC2_2  -.02
## 929     FAC3_2     FAC3_2
## 930     FAC3_2     FAC4_2  -.02
## 931     FAC4_2 Question_01  -.32
## 932     FAC4_2 Question_02  -.03
## 933     FAC4_2 Question_03  -.22
## 934     FAC4_2 Question_04  -.32
## 935     FAC4_2 Question_05  -.23
## 936     FAC4_2 Question_06  -.16
## 937     FAC4_2 Question_07  -.17
## 938     FAC4_2 Question_08  -.88
## 939     FAC4_2 Question_09  -.07
## 940     FAC4_2 Question_10  -.17
## 941     FAC4_2 Question_11  -.80
## 942     FAC4_2 Question_12  -.17
## 943     FAC4_2 Question_13  -.27
## 944     FAC4_2 Question_14  -.20
## 945     FAC4_2 Question_15  -.29
## 946     FAC4_2 Question_16  -.29
## 947     FAC4_2 Question_17  -.74
## 948     FAC4_2 Question_18  -.19
## 949     FAC4_2 Question_19  -.10
## 950     FAC4_2 Question_20  -.15
## 951     FAC4_2 Question_21  -.19
## 952     FAC4_2 Question_22   .10
## 953     FAC4_2 Question_23   .07
## 954     FAC4_2     FAC1_1  -.10
## 955     FAC4_2     FAC2_1  -.05
## 956     FAC4_2     FAC3_1 -1.00
## 957     FAC4_2     FAC4_1   .01
## 958     FAC4_2     FAC1_2  -.01
## 959     FAC4_2     FAC2_2   .00
## 960     FAC4_2     FAC3_2  -.02
## 961     FAC4_2     FAC4_2
```

```
library(psych)      #We are noe running parrallel analysis to give us a baseline for factor retention an
fa.parallel(df2_training[c(2:32)])
```

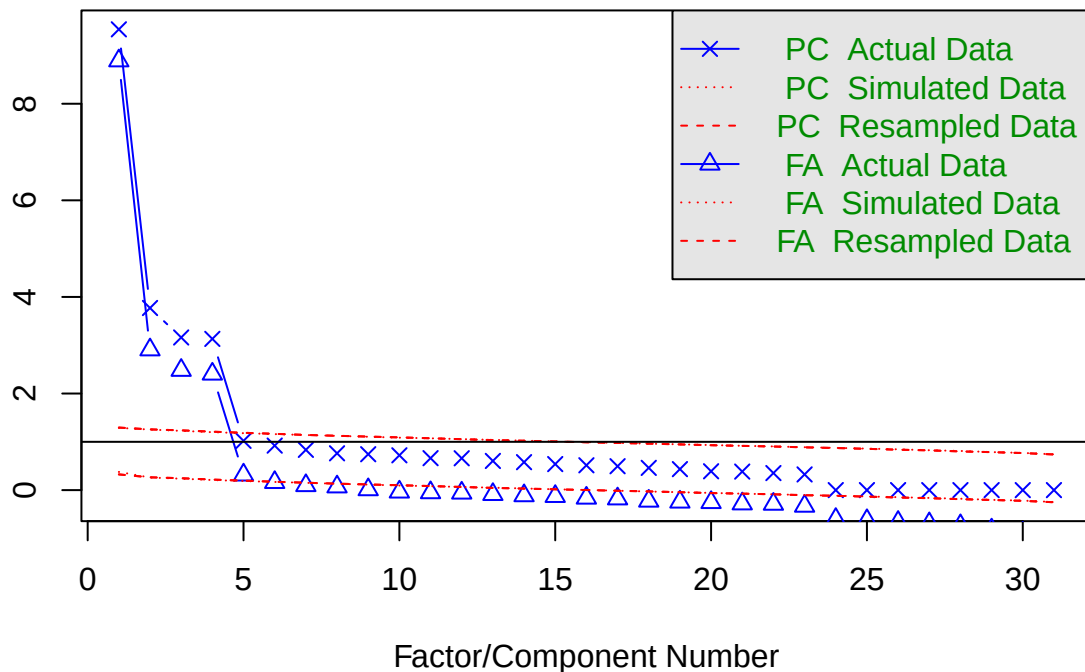
```
## In smc, smcs < 0 were set to .0
## In smc, smcs < 0 were set to .0
## In smc, smcs < 0 were set to .0
```

```
## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done
```

```
## In factor.scores, the correlation matrix is singular, the pseudo inverse is used
```

eigenvalues of principal components and factor analysis

Parallel Analysis Scree Plots



```
## Parallel analysis suggests that the number of factors = 5 and the number of components = 4
```

```
library(psych)
```

```
poly_cor_df2_training <- polychoric(df2_training[2:24])$rho
```

```
fa_pa_5_trn2 <- fa(poly_cor_df2_training, nfactors = 5,
```

```
  fm = "pa",
```

```
  #I am using Principal Axis Factoring because it doesn't assume strict normali
```

```
  rotate = "oblimin",
```

```
  #I am doing an oblimin rotation as this allows factors to corre
```

```
  n.obs = nrow(df2_training))
```

```
  #We will start with a 5 factor solution
```

```
print(fa_pa_5_trn2)
```

```
## Factor Analysis using method = pa
```

```
## Call: fa(r = poly_cor_df2_training, nfactors = 5, n.obs = nrow(df2_training),
```

```
##   rotate = "oblimin", fm = "pa")
```

```
## Standardized loadings (pattern matrix) based upon correlation matrix
```

```
##           PA3   PA5   PA1   PA2   PA4   h2   u2 com
```

```
## Question_01 -0.04  0.78  0.04 -0.03 -0.02 0.59 0.41 1.0
```

```
## Question_02 -0.04  0.02  0.00  0.54  0.15 0.36 0.64 1.2
```

```
## Question_03  0.00  0.25  0.08  0.42  0.21 0.52 0.48 2.3
```

```
## Question_04  0.09  0.53  0.09 -0.06  0.14 0.50 0.50 1.3
```

```
## Question_05  0.08  0.60  0.01  0.01 -0.03 0.41 0.59 1.0
```

```
## Question_06  0.84 -0.06  0.05  0.01 -0.13 0.63 0.37 1.1
```

```
## Question_07  0.59  0.04  0.00  0.03  0.23 0.56 0.44 1.3
```

```
## Question_08 -0.05  0.02  0.89 -0.08  0.00 0.75 0.25 1.0
```

```
## Question_09 -0.03 -0.01  0.10 -0.65  0.03 0.41 0.59 1.1
```

```
## Question_10  0.34  0.21  0.07  0.08 -0.16 0.26 0.74 2.5
```

```

## Question_11 0.07 -0.07 0.79 0.11 0.05 0.71 0.29 1.1
## Question_12 0.37 0.22 -0.01 0.06 0.25 0.50 0.50 2.5
## Question_13 0.57 0.09 0.10 0.01 0.06 0.52 0.48 1.1
## Question_14 0.42 0.22 0.00 0.10 0.13 0.49 0.51 1.9
## Question_15 0.28 0.17 0.15 0.18 0.01 0.37 0.63 3.1
## Question_16 0.04 0.58 0.04 0.18 0.04 0.54 0.46 1.2
## Question_17 0.09 0.17 0.63 -0.02 -0.03 0.61 0.39 1.2
## Question_18 0.61 0.09 0.02 0.00 0.17 0.59 0.41 1.2
## Question_19 0.14 -0.04 0.05 0.40 0.19 0.34 0.66 1.7
## Question_20 -0.09 -0.02 0.11 0.11 0.62 0.44 0.56 1.2
## Question_21 0.14 0.09 0.05 -0.02 0.68 0.68 0.32 1.1
## Question_22 -0.05 0.01 -0.08 -0.55 0.11 0.31 0.69 1.1
## Question_23 0.06 0.04 -0.09 -0.42 0.11 0.15 0.85 1.3
##
##
##          PA3  PA5  PA1  PA2  PA4
## SS loadings      2.96 2.51 2.26 1.91 1.58
## Proportion Var    0.13 0.11 0.10 0.08 0.07
## Cumulative Var    0.13 0.24 0.34 0.42 0.49
## Proportion Explained 0.26 0.22 0.20 0.17 0.14
## Cumulative Proportion 0.26 0.49 0.69 0.86 1.00
##
## With factor correlations of
##          PA3  PA5  PA1  PA2  PA4
## PA3 1.00 0.57 0.51 0.38 0.39
## PA5 0.57 1.00 0.58 0.30 0.52
## PA1 0.51 0.58 1.00 0.22 0.37
## PA2 0.38 0.30 0.22 1.00 0.36
## PA4 0.39 0.52 0.37 0.36 1.00
##
## Mean item complexity = 1.5
## Test of the hypothesis that 5 factors are sufficient.
##
## df null model = 253 with the objective function = 9.76 with Chi Square = 12449.72
## df of the model are 148 and the objective function was 0.57
##
## The root mean square of the residuals (RMSR) is 0.02
## The df corrected root mean square of the residuals is 0.02
##
## The harmonic n.obs is 1285 with the empirical chi square 190.25 with prob < 0.011
## The total n.obs was 1285 with Likelihood Chi Square = 724.72 with prob < 7.8e-77
##
## Tucker Lewis Index of factoring reliability = 0.919
## RMSEA index = 0.055 and the 90 % confidence intervals are 0.051 0.059
## BIC = -334.74
## Fit based upon off diagonal values = 1
## Measures of factor score adequacy
##
##          PA3  PA5  PA1  PA2  PA4
## Correlation of (regression) scores with factors 0.78 0.70 0.83 0.79 0.72
## Multiple R square of scores with factors 0.61 0.50 0.68 0.62 0.52
## Minimum correlation of possible factor scores 0.21 -0.01 0.36 0.24 0.03
print(fa_pa_5_trn2$loadings, cutoff = .3) # Input a cutoff to make this clearer to interpret
##
## Loadings:

```

```
##          PA3    PA5    PA1    PA2    PA4
## Question_01          0.784
## Question_02          0.537
## Question_03          0.423
## Question_04          0.532
## Question_05          0.600
## Question_06 0.836
## Question_07 0.591
## Question_08          0.887
## Question_09          -0.652
## Question_10 0.337
## Question_11          0.791
## Question_12 0.370
## Question_13 0.573
## Question_14 0.416
## Question_15
## Question_16          0.584
## Question_17          0.627
## Question_18 0.611
## Question_19          0.404
## Question_20          0.617
## Question_21          0.681
## Question_22          -0.554
## Question_23          -0.419
##
##          PA3    PA5    PA1    PA2    PA4
## SS loadings  2.340 1.904 1.908 1.661 1.202
## Proportion Var 0.102 0.083 0.083 0.072 0.052
## Cumulative Var 0.102 0.185 0.267 0.340 0.392
```

There is no cross loading here! This is great, as it tells us that we do not have to drop any items and can proceed with these 5 factors!

```
fa_pa_5_factor_loadings2 <- as.data.frame(round(unclass(fa_pa_5_trn2$loadings), 3)) %>%
  tibble::rownames_to_column("items") # "items" is what we want to call the column.
fa_pa_5_factor_loadings2          # I am now putting the factor loadings into spreadsheet format for interp
```

```
##          items    PA3    PA5    PA1    PA2    PA4
## 1 Question_01 -0.042 0.784 0.037 -0.028 -0.019
## 2 Question_02 -0.036 0.015 -0.001 0.537 0.151
## 3 Question_03 -0.001 0.249 0.082 0.423 0.210
## 4 Question_04 0.086 0.532 0.090 -0.060 0.143
## 5 Question_05 0.080 0.600 0.009 0.009 -0.035
## 6 Question_06 0.836 -0.063 0.049 0.010 -0.125
## 7 Question_07 0.591 0.045 -0.001 0.029 0.225
## 8 Question_08 -0.051 0.024 0.887 -0.077 0.003
## 9 Question_09 -0.025 -0.009 0.100 -0.652 0.032
## 10 Question_10 0.337 0.214 0.067 0.083 -0.161
## 11 Question_11 0.071 -0.067 0.791 0.114 0.047
## 12 Question_12 0.370 0.222 -0.014 0.064 0.249
## 13 Question_13 0.573 0.087 0.104 0.013 0.058
## 14 Question_14 0.416 0.224 0.005 0.101 0.126
## 15 Question_15 0.281 0.167 0.155 0.184 0.005
## 16 Question_16 0.041 0.584 0.044 0.176 0.040
## 17 Question_17 0.094 0.172 0.627 -0.020 -0.026
```

```
## 18 Question_18 0.611 0.093 0.020 0.000 0.167
## 19 Question_19 0.139 -0.039 0.047 0.404 0.190
## 20 Question_20 -0.092 -0.022 0.114 0.113 0.617
## 21 Question_21 0.143 0.094 0.047 -0.017 0.681
## 22 Question_22 -0.050 0.009 -0.083 -0.554 0.114
## 23 Question_23 0.056 0.038 -0.088 -0.419 0.110

summary(fa_pa_5_trn2,
        fit.measures = TRUE,
        standardized = TRUE)      #I am running this chunk to get key fit measures for the EFA

##
## Factor analysis with Call: fa(r = poly_cor_df2_training, nfactors = 5, n.obs = nrow(df2_training),
##   rotate = "oblimin", fm = "pa")
##
## Test of the hypothesis that 5 factors are sufficient.
## The degrees of freedom for the model is 148 and the objective function was 0.57
## The number of observations was 1285 with Chi Square = 724.72 with prob < 7.8e-77
##
## The root mean square of the residuals (RMSA) is 0.02
## The df corrected root mean square of the residuals is 0.02
##
## Tucker Lewis Index of factoring reliability = 0.919
## RMSEA index = 0.055 and the 90 % confidence intervals are 0.051 0.059
## BIC = -334.74
## With factor correlations of
##   PA3 PA5 PA1 PA2 PA4
## PA3 1.00 0.57 0.51 0.38 0.39
## PA5 0.57 1.00 0.58 0.30 0.52
## PA1 0.51 0.58 1.00 0.22 0.37
## PA2 0.38 0.30 0.22 1.00 0.36
## PA4 0.39 0.52 0.37 0.36 1.00
```

The TLI of my final model is 0.919. This is higher than 0.90, therefore, this is an acceptable TLI. The RMSEA of my final model is 0.055 which is almost excellent as it is almost under or equal to 0.05. But since it is just over 0.05, I will refer to this as acceptable at a 90% confidence interval as it is still below 0.08.

#Scale Analysis

```
library(dplyr)
df2_items <- df2_training %>%
  dplyr::select(-c(FAC1_1, FAC2_1, FAC3_1, FAC4_1, FAC1_2, FAC2_2, FAC3_2, FAC4_2)) #I am now cleaning

df2_keys_list <- list(computers = c(7, 8, 11, 13, 14, 15, 19),
                      statistics = c(2, 5, 6, 17),
                      mathematics = c(9, 12, 18),
                      SPSS = c(3, 4, 10, 20, 23, 24),
                      sleep = c(21, 22)
                      )

df2_keys <- make.keys(df2_items, df2_keys_list, item.labels = colnames(df2_items))

scores <- scoreItems(df2_keys, df2_items, impute = "none",
                     min = 1, max = 6, digits = 3) #Now I will score the items

## Warning in sqrt(y[i] * y[j]): NaNs produced
## Warning in sqrt(y[i] * y[j]): NaNs produced
```

```
## Warning in sqrt(y[i] * y[j]): NaNs produced
## Warning in sqrt(y[i] * y[j]): NaNs produced
```

```
head(scores$scores)
```

```
##      computers statistics mathematics      SPSS sleep
## [1,]  3.142857      3.25    2.000000 3.833333   4.0
## [2,]  2.714286      2.00    2.000000 3.833333   3.0
## [3,]  2.000000      2.00    2.000000 3.333333   2.0
## [4,]  2.142857      2.00    2.000000 3.500000   2.5
## [5,]  1.857143      2.00    1.666667 3.666667   2.5
## [6,]  3.142857      2.50    2.000000 3.333333   5.0
```

```
scores_df2 <- as.data.frame(scores$scores)
```

```
#Scale Reliability of Computers
```

```
COMP_ALPHA <- psych::alpha(df2_items[, c(7, 8, 11, 13, 14, 15, 19)]) #I am using `alpha` from the `psych` package
```

```
COMP_total <- round(as.data.frame(COMP_ALPHA$total), 3)
COMP_alpha_drop <- round(as.data.frame(COMP_ALPHA$alpha.drop), 3)
COMP_item_stat <- round(as.data.frame(COMP_ALPHA$item.stats), 3)
```

```
COMP_ALPHA
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df2_items[, c(7, 8, 11, 13, 14, 15, 19)])
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean   sd median_r
##      0.83      0.83      0.82      0.41 4.9 0.007   2.7 0.71      0.45
##
##      95% confidence boundaries
##      lower alpha upper
## Feldt      0.82 0.83 0.85
## Duhachek    0.82 0.83 0.85
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se  var.r med.r
## Question_06      0.81      0.81      0.78      0.41 4.2 0.0081 0.0104 0.45
## Question_07      0.80      0.80      0.78      0.40 4.1 0.0083 0.0106 0.45
## Question_10      0.84      0.84      0.82      0.47 5.2 0.0070 0.0026 0.47
## Question_12      0.81      0.81      0.79      0.41 4.2 0.0080 0.0096 0.45
## Question_13      0.80      0.80      0.78      0.40 4.0 0.0083 0.0099 0.44
## Question_14      0.81      0.81      0.79      0.41 4.2 0.0080 0.0100 0.44
## Question_18      0.79      0.79      0.77      0.39 3.8 0.0087 0.0085 0.43
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean   sd
## Question_06 1285 0.73 0.72 0.66 0.60 2.3 1.14
## Question_07 1285 0.75 0.73 0.68 0.62 2.9 1.10
## Question_10 1285 0.53 0.55 0.41 0.38 2.3 0.87
## Question_12 1285 0.70 0.71 0.64 0.58 3.1 0.92
## Question_13 1285 0.74 0.74 0.69 0.63 2.5 0.94
## Question_14 1285 0.71 0.71 0.64 0.58 2.9 1.00
## Question_18 1285 0.78 0.78 0.74 0.68 2.6 1.05
##
```



```
## Non missing response frequency for each item
##      1      2      3      4      5 miss
## Question_06 0.26 0.44 0.13 0.11 0.06    0
## Question_07 0.07 0.34 0.26 0.24 0.09    0
## Question_10 0.13 0.57 0.19 0.10 0.02    0
## Question_12 0.02 0.20 0.47 0.22 0.09    0
## Question_13 0.12 0.48 0.25 0.13 0.02    0
## Question_14 0.06 0.32 0.37 0.18 0.07    0
## Question_18 0.14 0.36 0.31 0.13 0.05    0

#Scale Reliability of Statistics
STAT_ALPHA <- psych::alpha(df2_items[, c(2, 5, 6, 17)]) #I am using `alpha` from the `psych` package r

STAT_total <- round(as.data.frame(STAT_ALPHA$total), 3)
STAT_alpha_drop <- round(as.data.frame(STAT_ALPHA$alpha.drop), 3)
STAT_item_stat <- round(as.data.frame(STAT_ALPHA$item.stats), 3)

STAT_ALPHA

##
## Reliability analysis
## Call: psych::alpha(x = df2_items[, c(2, 5, 6, 17)])
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.75      0.76      0.7      0.44 3.1 0.011  2.7 0.7      0.42
##
##      95% confidence boundaries
##              lower alpha upper
## Feldt      0.73  0.75  0.77
## Duhachek  0.73  0.75  0.78
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se   var.r med.r
## Question_01      0.67      0.67  0.58      0.40 2.0   0.016 0.00016  0.41
## Question_04      0.70      0.70  0.61      0.44 2.3   0.015 0.00368  0.42
## Question_05      0.72      0.72  0.64      0.47 2.6   0.014 0.00226  0.48
## Question_16      0.70      0.70  0.61      0.43 2.3   0.015 0.00135  0.42
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## Question_01 1285 0.78 0.79 0.70  0.60 2.4 0.85
## Question_04 1285 0.76 0.76 0.63  0.55 2.8 0.94
## Question_05 1285 0.74 0.73 0.58  0.51 2.7 0.96
## Question_16 1285 0.76 0.76 0.64  0.55 2.9 0.92
##
## Non missing response frequency for each item
##      1      2      3      4      5 miss
## Question_01 0.11 0.52 0.27 0.08 0.02    0
## Question_04 0.05 0.39 0.35 0.16 0.05    0
## Question_05 0.06 0.43 0.28 0.20 0.04    0
## Question_16 0.04 0.32 0.41 0.17 0.05    0

#Scale Reliability of Mathematics
```

```
MATH_ALPHA <- psych::alpha(df2_items[, c(9, 12, 18)]) #I am using `alpha` from the `psych` package rat
```

```
MATH_total <- round(as.data.frame(MATH_ALPHA$total), 3)
MATH_alpha_drop <- round(as.data.frame(MATH_ALPHA$alpha.drop), 3)
MATH_item_stat <- round(as.data.frame(MATH_ALPHA$item.stats), 3)
```

```
MATH_ALPHA
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df2_items[, c(9, 12, 18)])
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean   sd median_r
##      0.81      0.81      0.75      0.59 4.4 0.009   2.3 0.75      0.58
##
##      95% confidence boundaries
##              lower alpha upper
## Feldt      0.8  0.81  0.83
## Duhachek    0.8  0.81  0.83
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## Question_08    0.72    0.72    0.57    0.57 2.6   0.015   NA  0.57
## Question_11    0.73    0.74    0.58    0.58 2.8   0.015   NA  0.58
## Question_17    0.77    0.77    0.63    0.63 3.4   0.013   NA  0.63
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean   sd
## Question_08 1285  0.86  0.86  0.76   0.68  2.2 0.86
## Question_11 1285  0.86  0.86  0.75   0.67  2.3 0.88
## Question_17 1285  0.84  0.84  0.70   0.64  2.5 0.90
##
## Non missing response frequency for each item
##      1  2  3  4  5 miss
## Question_08 0.14 0.57 0.21 0.05 0.03  0
## Question_11 0.16 0.52 0.23 0.07 0.02  0
## Question_17 0.09 0.51 0.27 0.10 0.03  0
```

```
#Scale Reliability of SPSS
```

```
SPSS_ALPHA <- psych::alpha(df2_items[, c(3, 4, 10, 20, 23, 24)]) #I am using `alpha` from the `psych`
```

```
## Warning in psych::alpha(df2_items[, c(3, 4, 10, 20, 23, 24)]): Some items were negatively correlated
## should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' option
## Some items ( Question_09 Question_22 Question_23 ) were negatively correlated with the first principal
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' option
```

```
SPSS_total <- round(as.data.frame(SPSS_ALPHA$total), 3)
SPSS_alpha_drop <- round(as.data.frame(SPSS_ALPHA$alpha.drop), 3)
SPSS_item_stat <- round(as.data.frame(SPSS_ALPHA$item.stats), 3)
```

```
SPSS_ALPHA
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df2_items[, c(3, 4, 10, 20, 23, 24)])
##
##      raw_alpha std.alpha G6(smc) average_r   S/N   ase mean   sd median_r
##      -0.24      -0.21   0.033      -0.03 -0.17 0.055   3.4 0.4      -0.14
##
##      95% confidence boundaries
##              lower alpha upper
## Feldt      -0.35 -0.24 -0.14
## Duhachek -0.35 -0.24 -0.13
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r   S/N alpha se var.r med.r
## Question_02   -0.212   -0.200   0.028   -0.0345 -0.167   0.055 0.074 -0.16
## Question_03   -0.189   -0.225  -0.015   -0.0381 -0.184   0.052 0.062 -0.13
## Question_09   -0.022   -0.013   0.144   -0.0026 -0.013   0.045 0.068 -0.13
## Question_19   -0.185   -0.206   0.020   -0.0353 -0.171   0.052 0.072 -0.15
## Question_22   -0.202   -0.113   0.086   -0.0208 -0.102   0.054 0.075 -0.13
## Question_23   -0.362   -0.288  -0.002   -0.0469 -0.224   0.062 0.091 -0.24
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean   sd
## Question_02 1285  0.30  0.40  0.18 -0.064  4.4 0.85
## Question_03 1285  0.37  0.41  0.42 -0.082  3.4 1.07
## Question_09 1285  0.36  0.26 -0.52 -0.182  2.8 1.27
## Question_19 1285  0.38  0.40  0.22 -0.084  3.7 1.10
## Question_22 1285  0.36  0.34 -0.16 -0.073  2.8 1.03
## Question_23 1285  0.47  0.45  0.27  0.037  3.4 1.03
##
## Non missing response frequency for each item
##      1 2 3 4 5 miss
## Question_02 0.01 0.04 0.07 0.32 0.56 0
## Question_03 0.03 0.18 0.35 0.26 0.19 0
## Question_09 0.20 0.22 0.21 0.29 0.08 0
## Question_19 0.02 0.15 0.20 0.34 0.29 0
## Question_22 0.10 0.27 0.33 0.26 0.04 0
## Question_23 0.06 0.13 0.28 0.42 0.11 0
```

#Scale Reliability of Sleep

```
SLEEP_ALPHA <- psych::alpha(df2_items[, c(21, 22)]) #I am using `alpha` from the `psych` package rather
```

```
SLEEP_total <- round(as.data.frame(SLEEP_ALPHA$total), 3)
SLEEP_alpha_drop <- round(as.data.frame(SLEEP_ALPHA$alpha.drop), 3)
SLEEP_item_stat <- round(as.data.frame(SLEEP_ALPHA$item.stats), 3)
```

```
SLEEP_ALPHA
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df2_items[, c(21, 22)])
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean   sd median_r
##      0.66      0.66      0.5      0.5 2 0.019 3.4 0.88      0.5
```

```

##
##      95% confidence boundaries
##      lower alpha upper
## Feldt      0.62 0.66 0.7
## Duhachek   0.63 0.66 0.7
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## Question_20      0.53      0.5 0.25      0.5 0.99      NA      0 0.5
## Question_21      0.47      0.5 0.25      0.5 0.99      NA      0 0.5
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean sd
## Question_20 1285 0.87 0.87 0.61 0.5 3.6 1.05
## Question_21 1285 0.86 0.87 0.61 0.5 3.1 0.98
##
## Non missing response frequency for each item
##      1 2 3 4 5 miss
## Question_20 0.02 0.14 0.25 0.37 0.21 0
## Question_21 0.02 0.26 0.35 0.27 0.09 0

```